



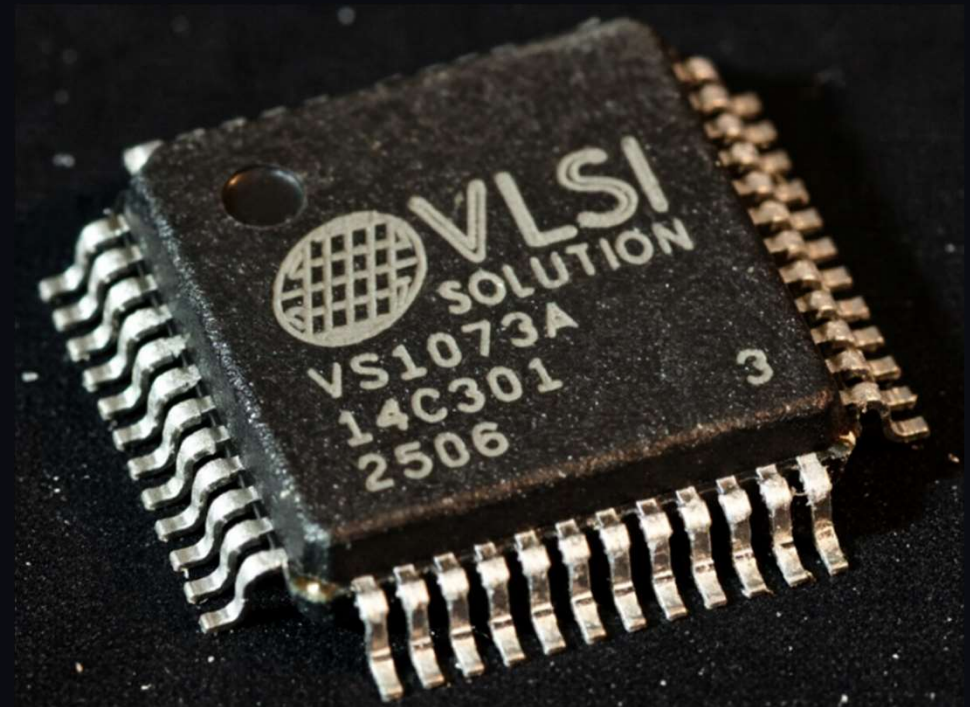
VLSI Design

Dr. Rizki Ardianto Priramadhi, S.T., M.T.
Telkom University,
Indonesia

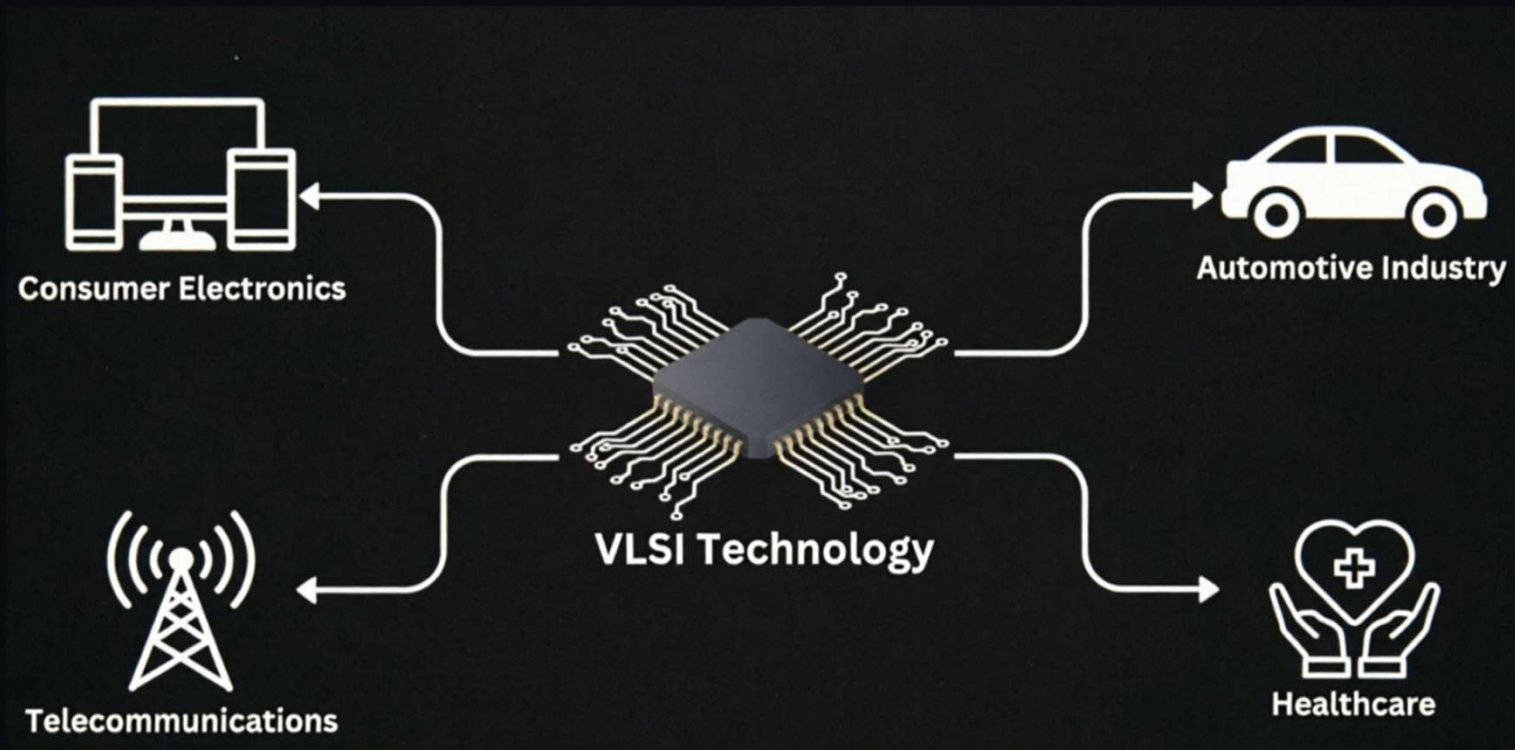
Very Large Scale Integration (VLSI)

VLSI Technology

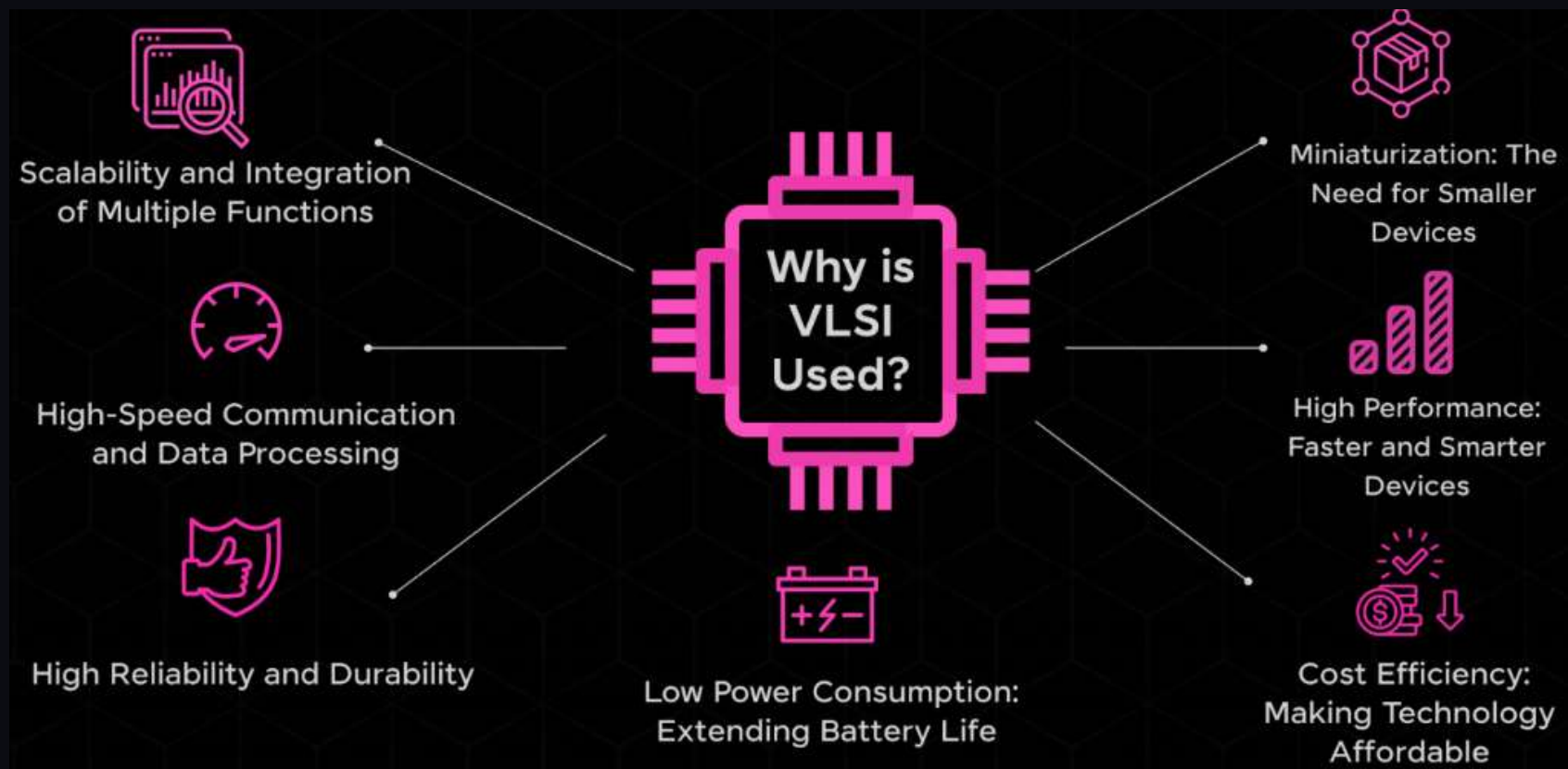
- Process of integrating hundreds of thousands of transistors onto a singular silicon semiconductor microchip.
- Dates back to the late 1970s when advanced level processor (computer) microchips were also in their development stages.
- Two of the most common VLSI devices are Processor and Microcontroller.
- VLSI = Integrated Circuit



VLSI in Modern Life

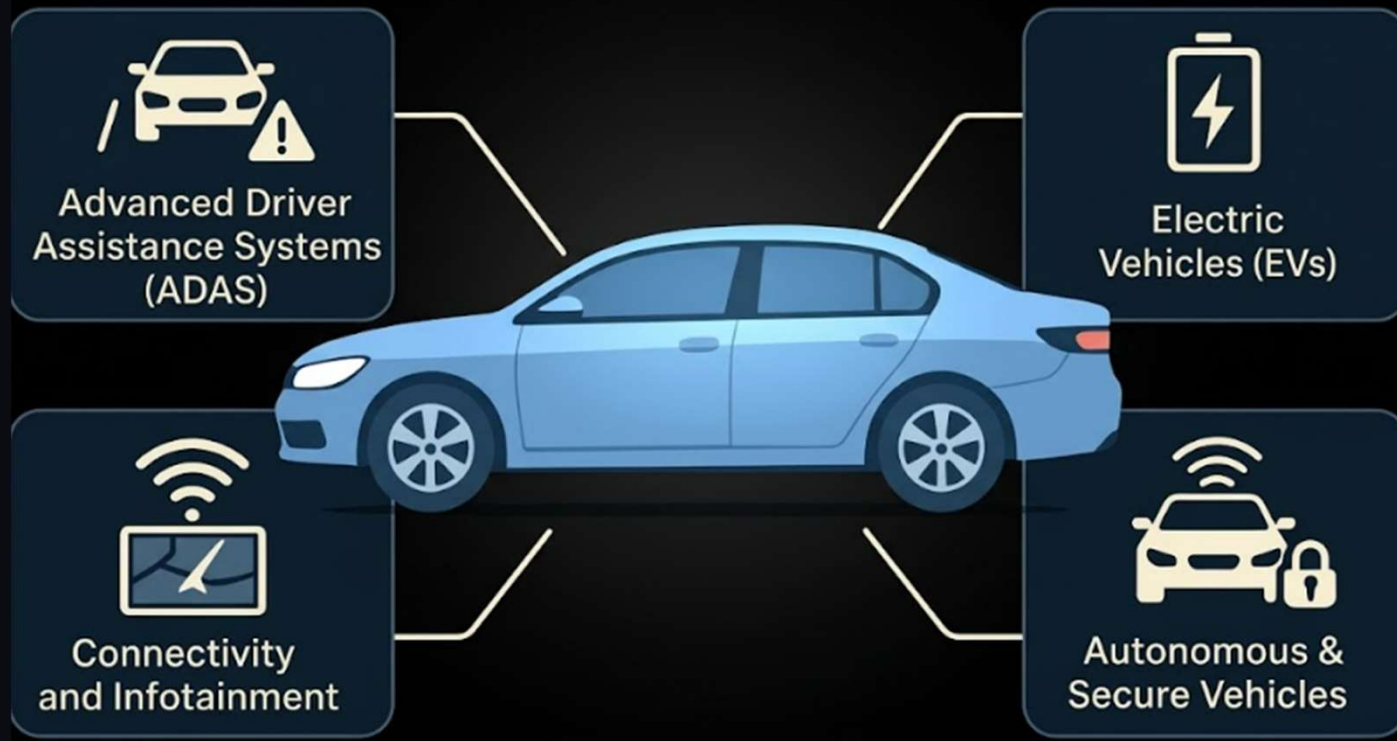


VLSI in Wireless Communication

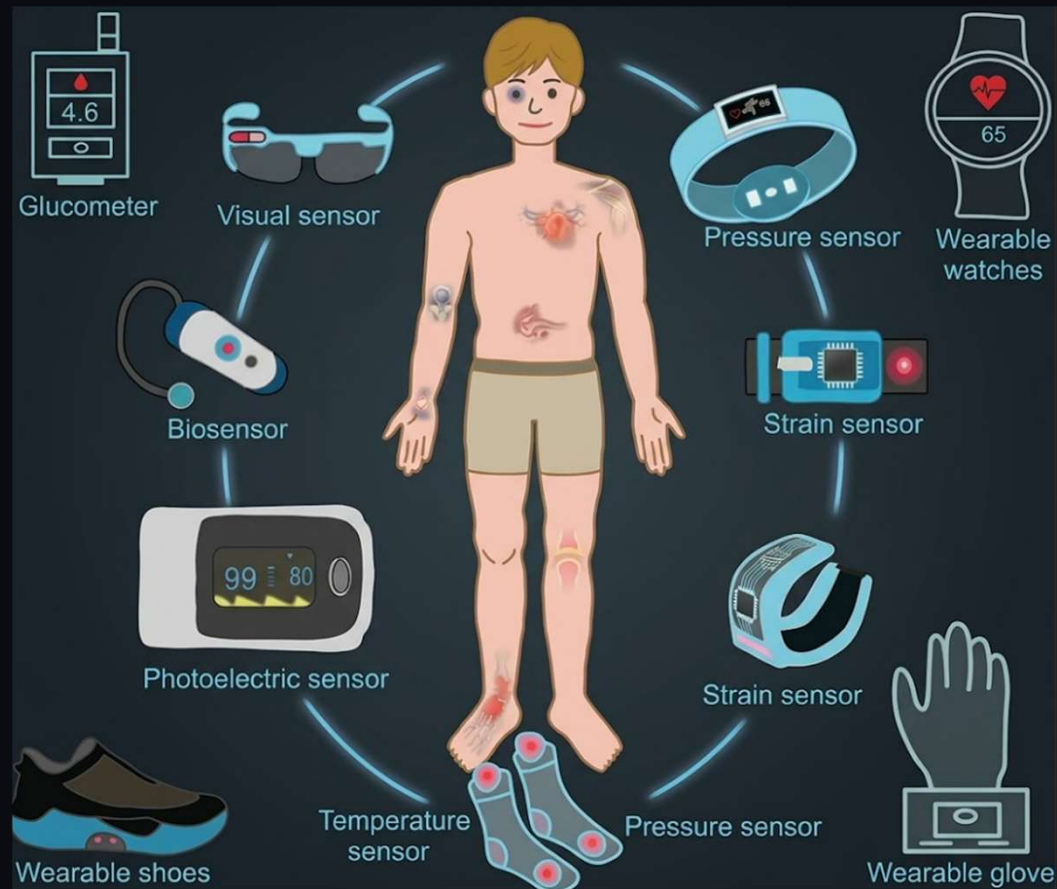


VLSI in Automotive Technology

Driving the Future of Smart Vehicles



VLSI in Medical System



History of VLSI : First Transistor

"Study the past if you would define the future."

The **first working transistor** was the point-contact transistor, invented in **December 1947** by physicists **John Bardeen, Walter Brattain, and William Shockley** at [Bell Labs](#).

~ 1.3 cm tall

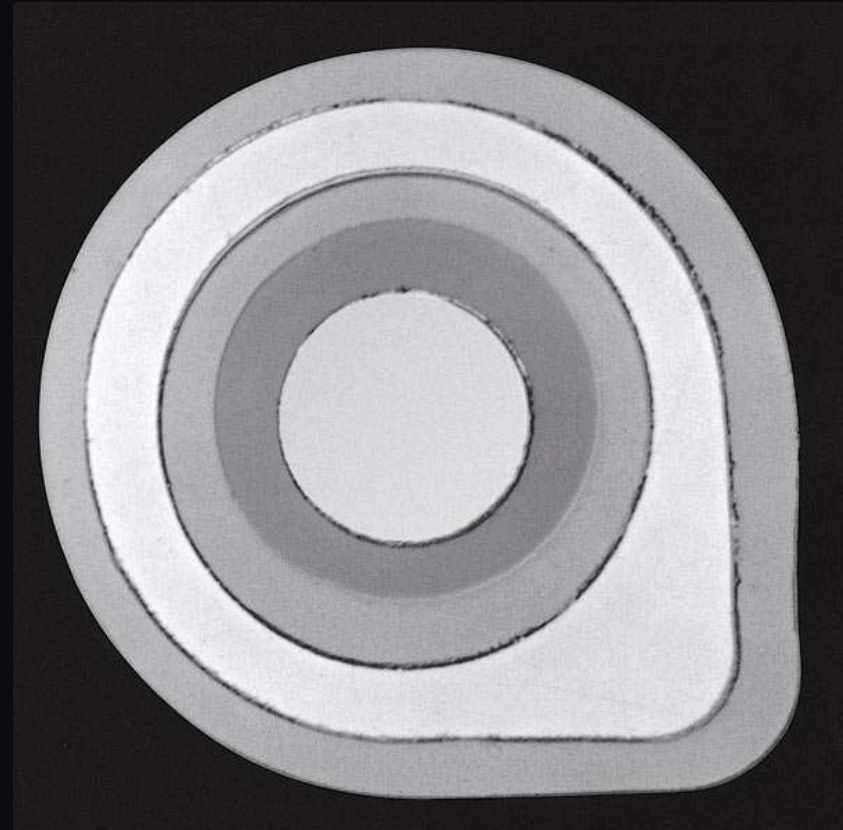




History of VLSI : First Flat-Transistor

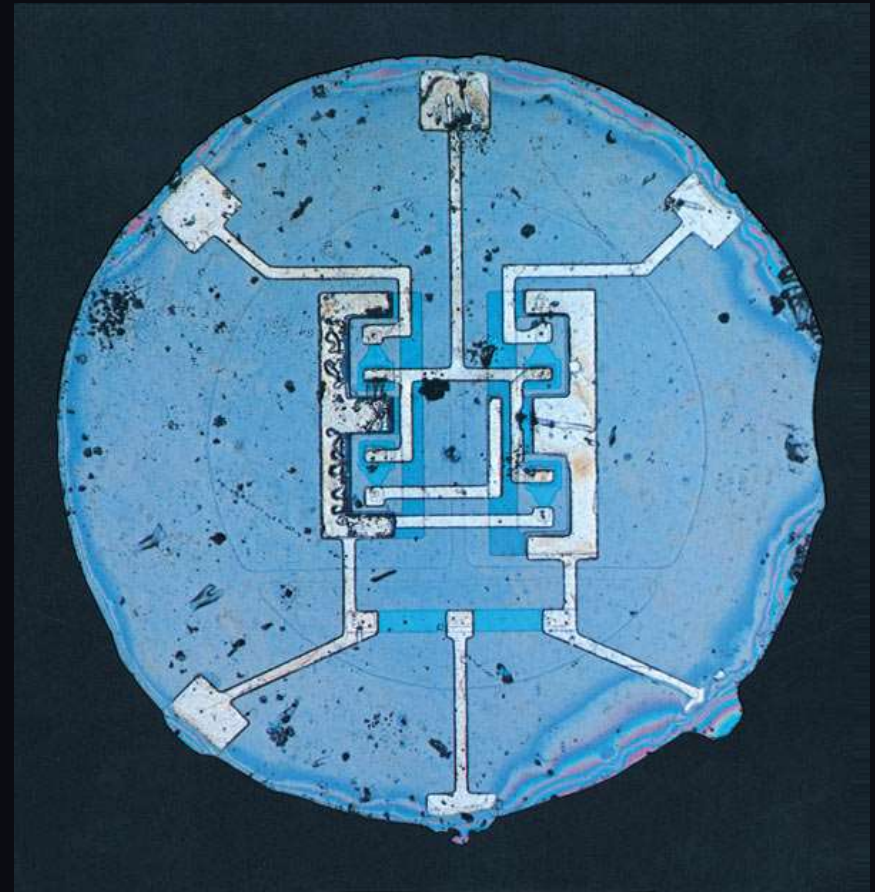
Jean Hoerni, a cofounder of Fairchild Semiconductor, invented the first planar, or flat, transistor in 1959. His novel manufacturing approach was to directly imprint semiconducting and insulating channels onto a π -silicon wafer. The process left intact a protective layer of silicon dioxide that formed naturally on top of the wafer and prevented contamination. The result was the best-performing transistor of its time.

(1.27 x 1.27 mm)



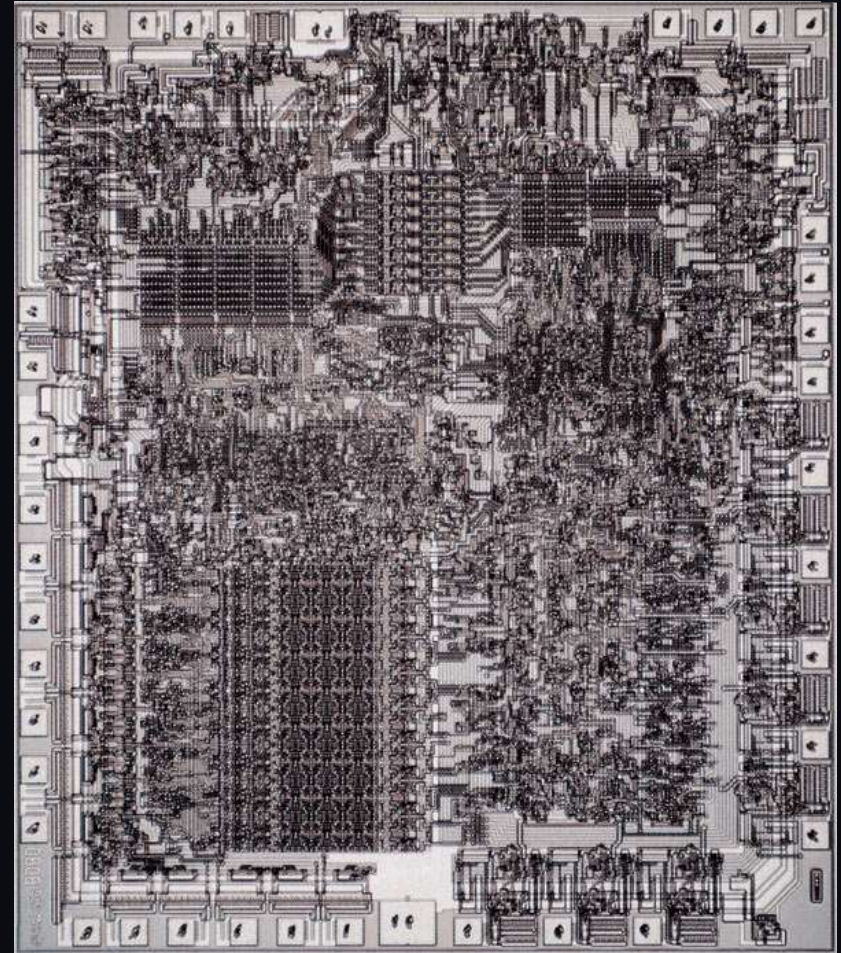
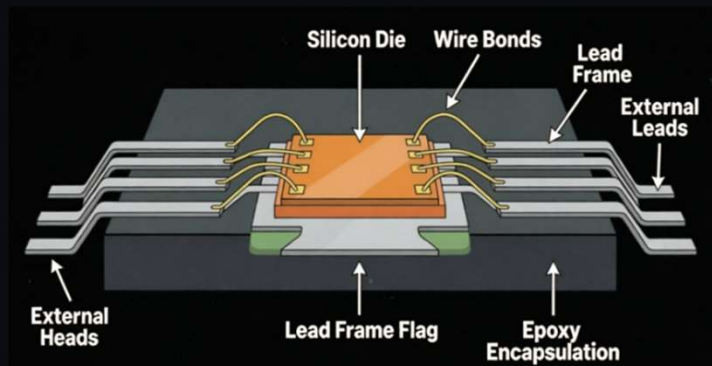
History of VLSI : First Multi-Transistor

- Robert Noyce — in 1961. This resistor-transistor logic chip was one of the first commercial integrated circuits. It has four transistors (quadrants in the middle). The white lines are metal traces, which connect the transistors to the two resistors below (horizontal blue bar). The Apollo Guidance Computer used the chip.
- Size : 3.18mm x 3.18mm
- Transistor count : 4



History of VLSI : First (micro)Processor

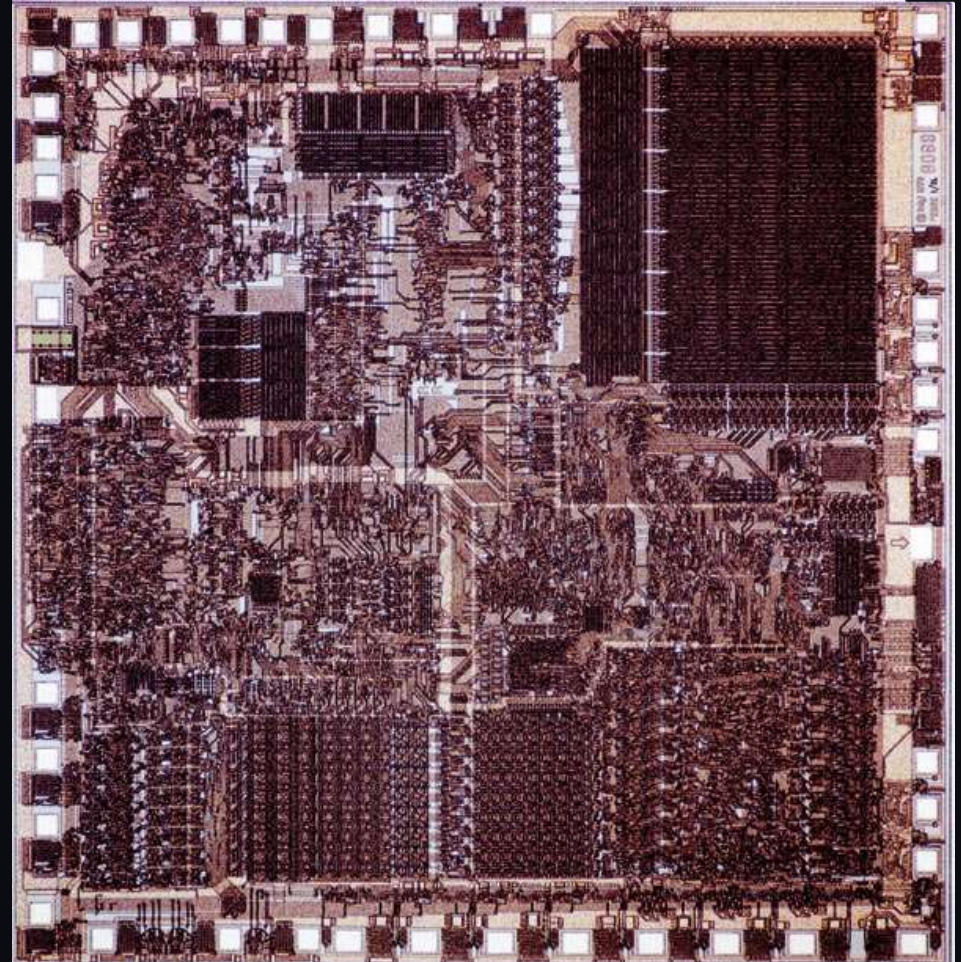
- In 1974, Intel introduced the 8080.
- Roughly 5,000 transistors.
- Speed : 2 MHz
- Size : 20 mm²
- the heart of the Altair personal computer.



Silicon die of Intel 8080

History of VLSI : First Processor

- In 1978 Intel's 8086 chip, established the x86 architecture
- 29,000 transistors,
- Speed : 5-10 MHz
- Size : 33 mm²

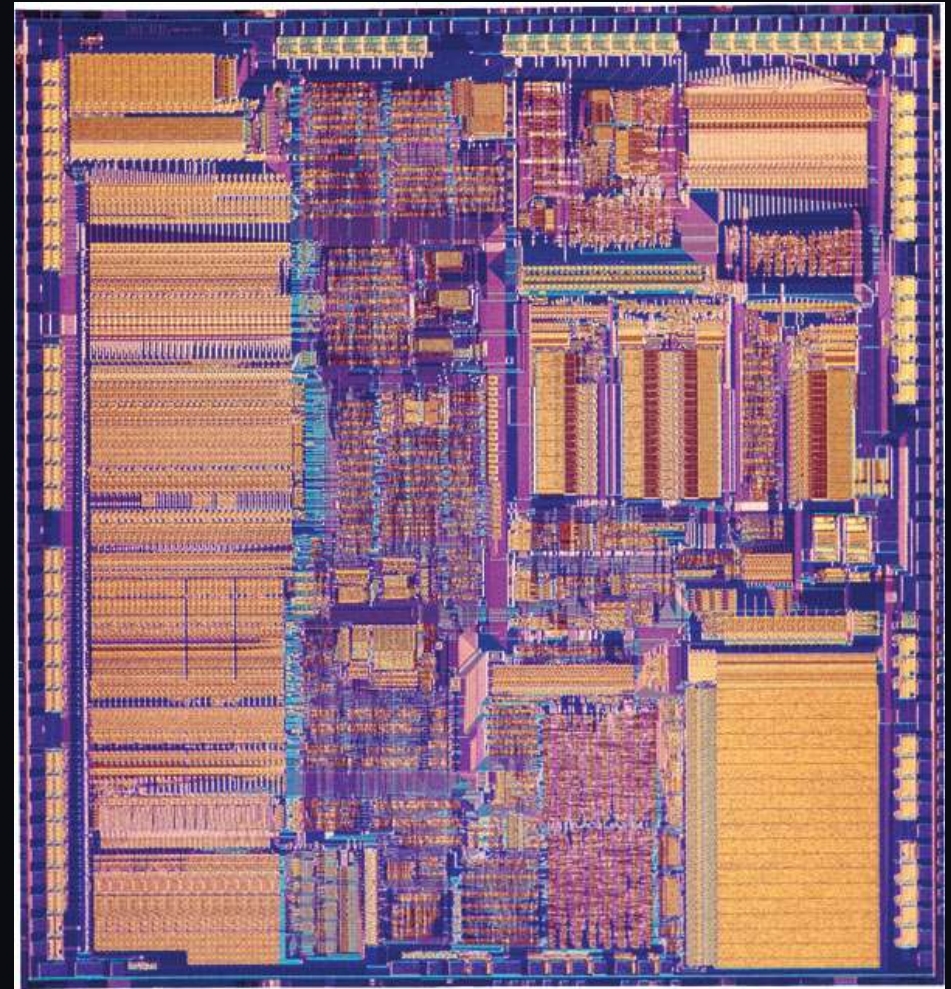


Silicon die photos of Intel 8086



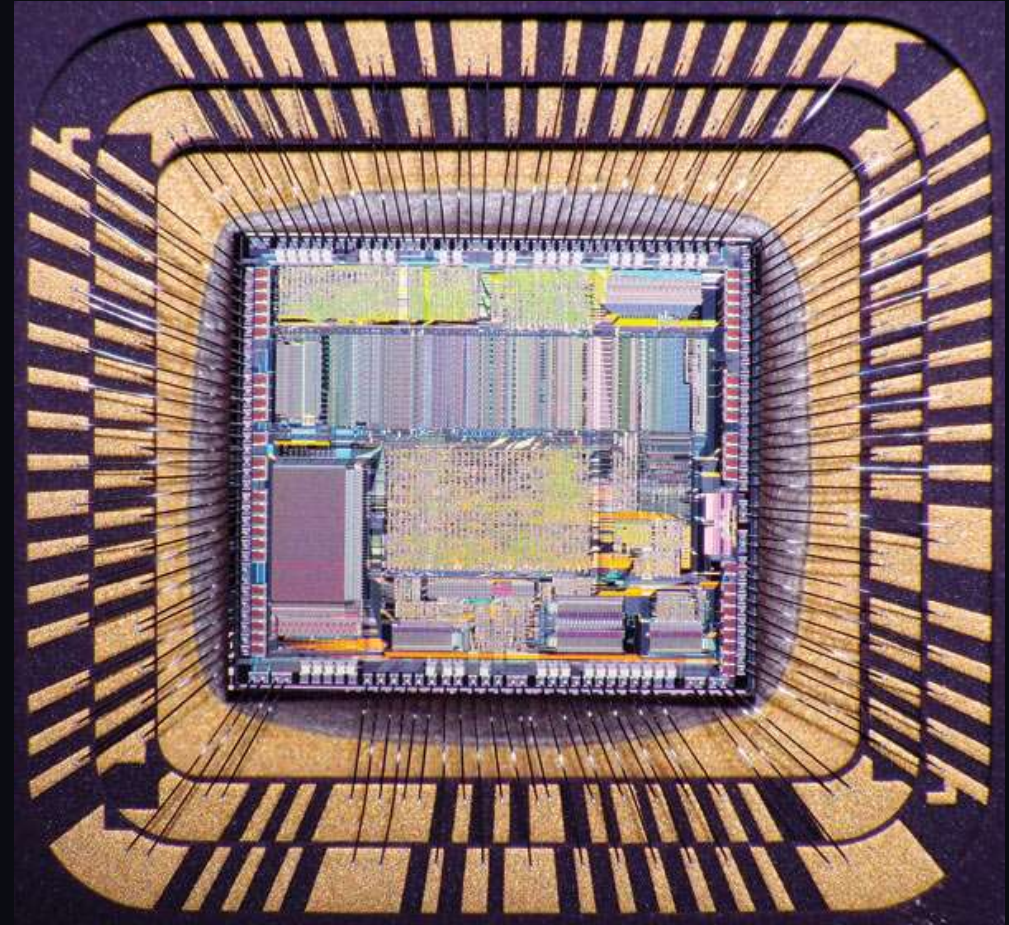
History of VLSI : i386

- Intel's 386, released in 1985, allowed a computer to work on multiple applications at the same time.
- Transistors Count : 275,000
- Speed 12-33 MHz
- Size : 104 mm² (1,5μm) ; 39 mm² (1μm))



History of VLSI : AMD Processor

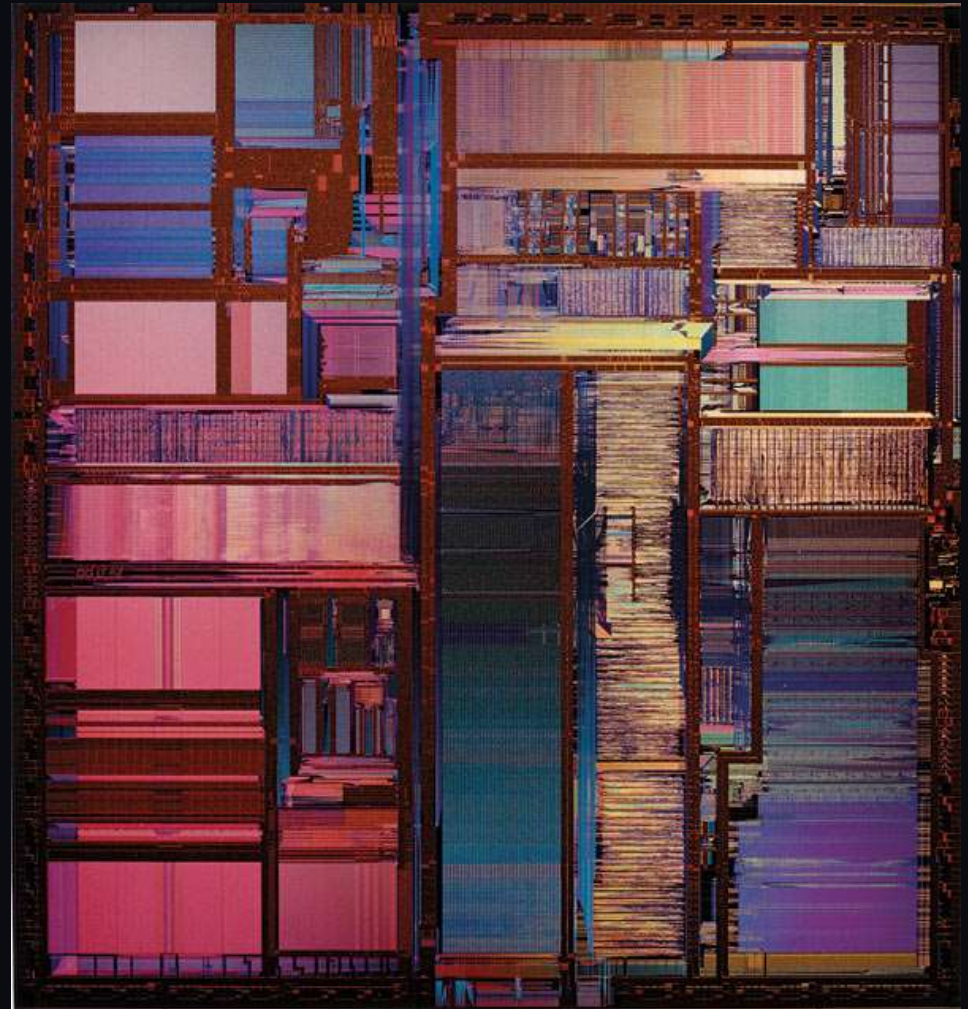
- In 1991, AMD released its own 386 microprocessor
- Transistor Count : ~ 275,000 transistors
- Size : ~36.8 mm × 36.8 mm
- Speed : 40 Mhz





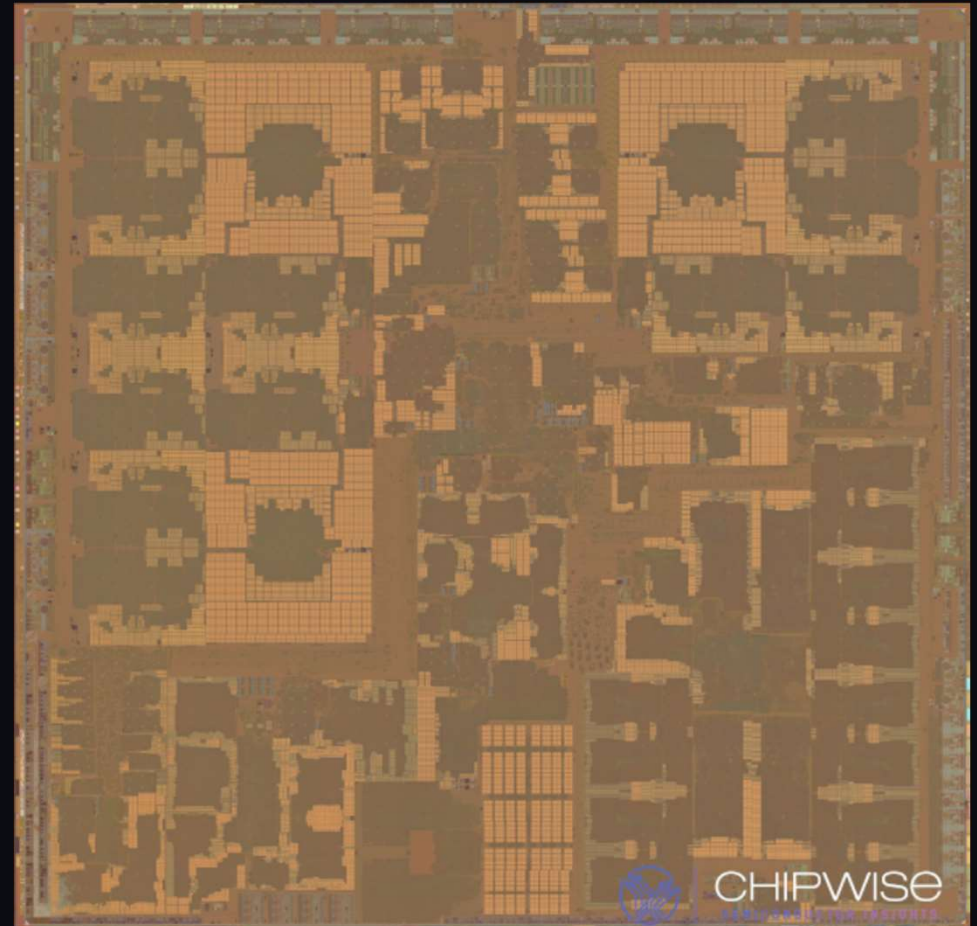
History of VLSI : Intel Pentium

- The Pentium processor debuted in 1993
- Transistor Count : 3.1 million transistors.
- Size : 16.7×17.6 mm
- Speed : 66 MHz



Today VLSI : ARM SnapDragon

- Snapdragon X elite, 2023
- Transistor Count : ~27 Billion
- Size : 170 mm²
- Speed : 3.8 GHz
- 4 nm process
- Power : 22 - 82 watt



Today VLSI : Apple - ARM

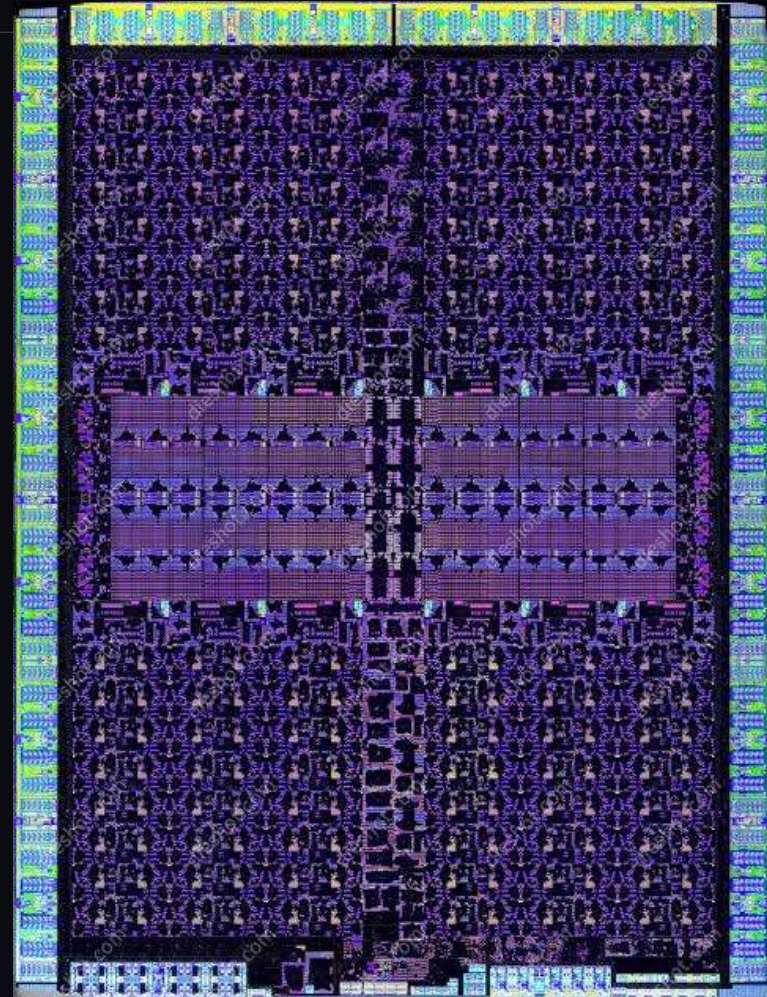
- Apple M4, 2024
- Transistors Count : 28 billion
- Size : 167 mm²
- Speed : 2.6 - 4.5 GHz
- 3 nm process
- Power : 40 watt





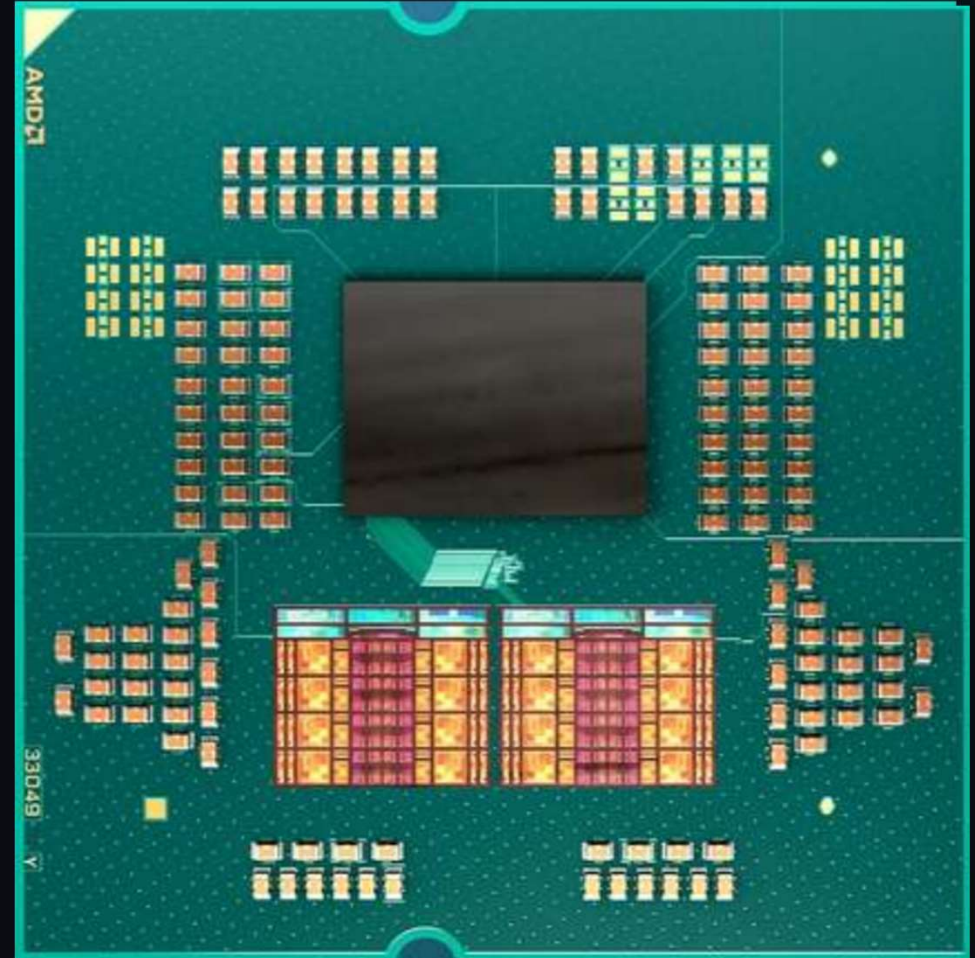
Today's VLSI : GPU

- Nvidia GeForce RTX 5090, 2025
- Transistor Count : 92.2 billion
- Size : 750 mm²
- Speed : 2.4 GHz
- 4 nm process (custom)
- Power : 400 – 550 watt



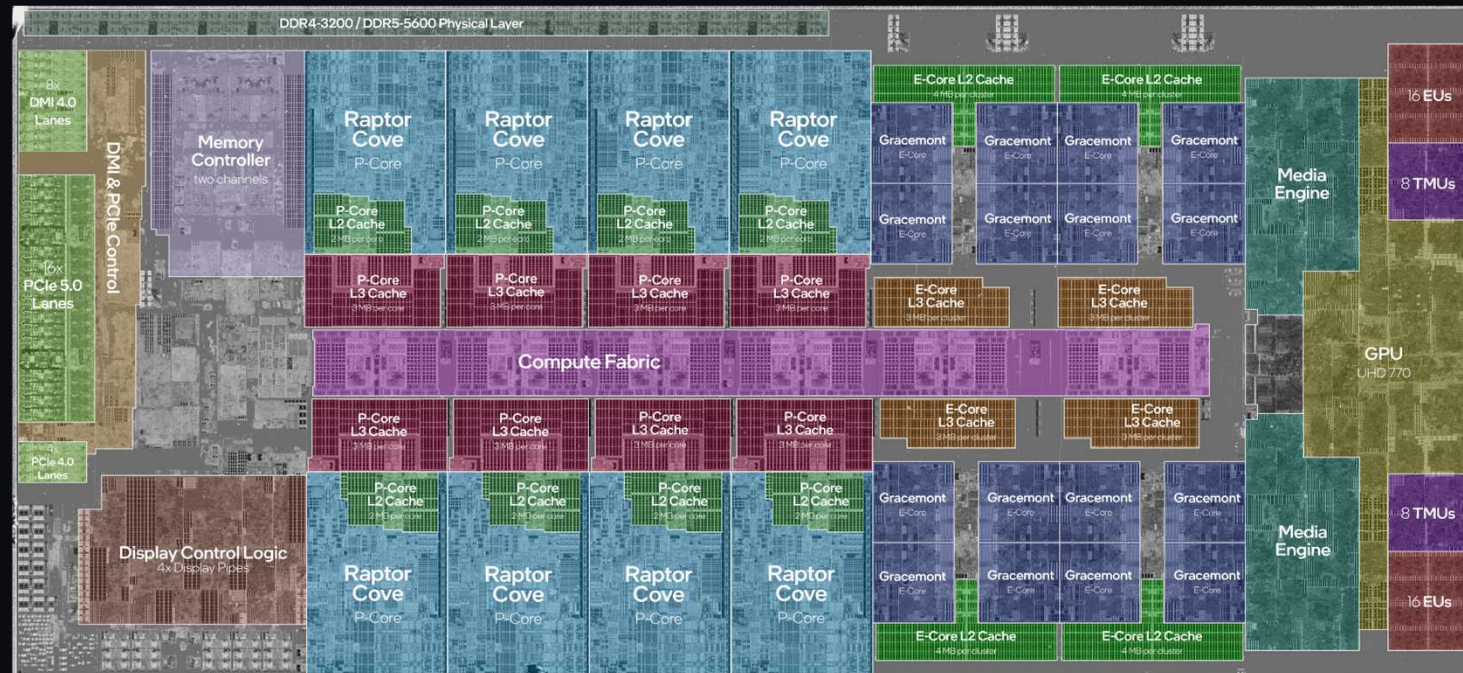
Today's VLSI : AMD

- AMD Ryzen 9 9950x, 2024
- Transistor Count : 20.6 Billion
- Size : ~263mm²
- Speed : 4.3 – 5.7 GHz
- 4 & 6 nm Process
- Power : 100 - 130 watt

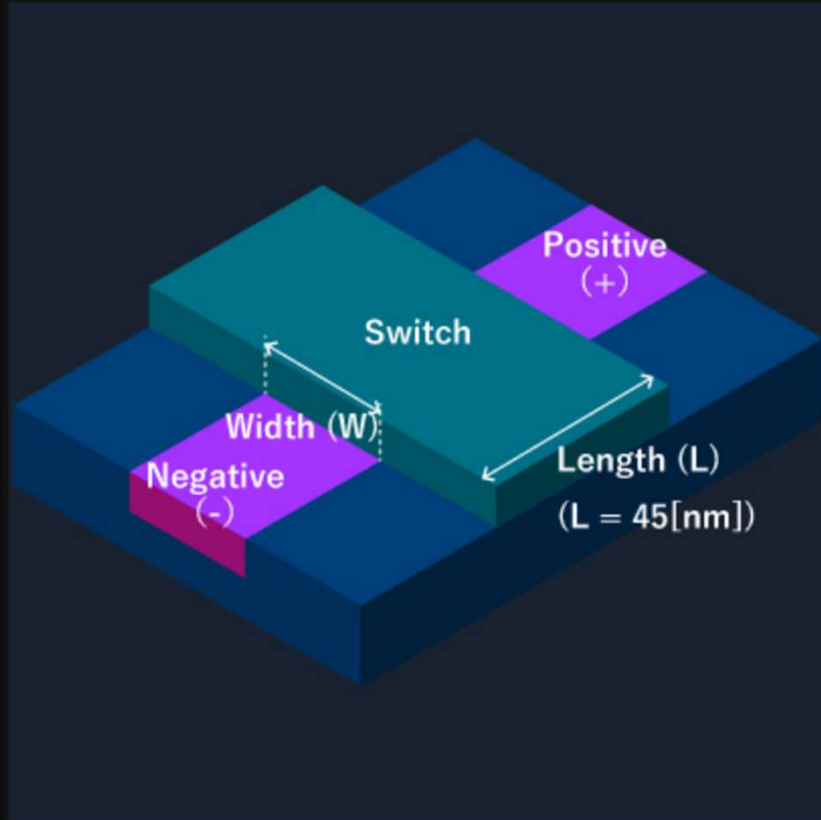
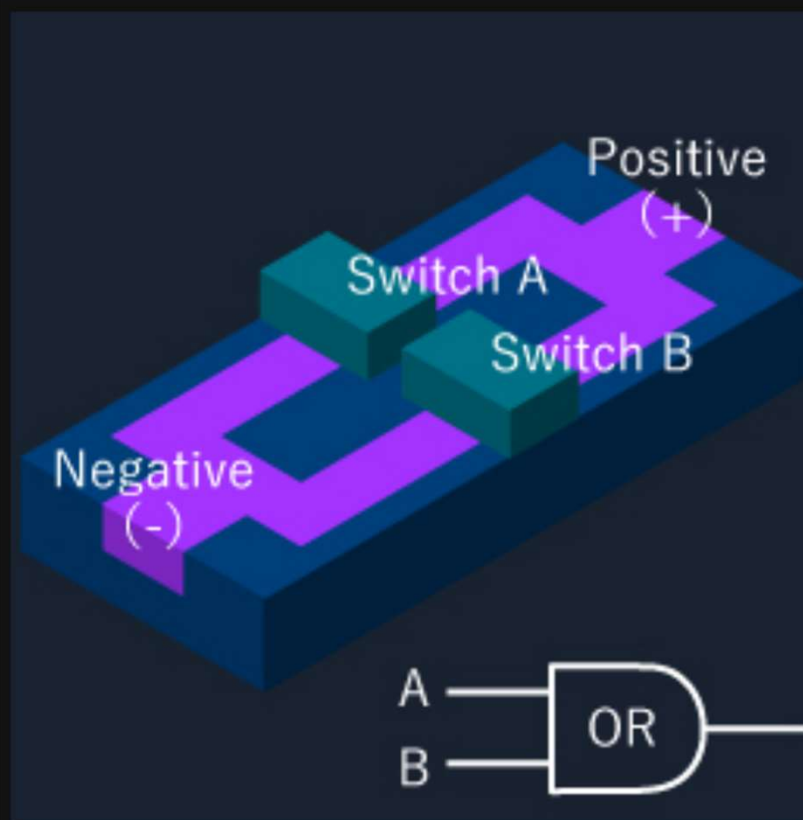


Today's VLSI : Intel i9

- Intel i9 14900K, 2023
- Transistor Count : 25.9 Billion
- Size : 257 mm²
- Speed : 2.4 – 6 GHz
- 10nm process
- Power : 140 – 160 Watt



Process Size Means (nm)



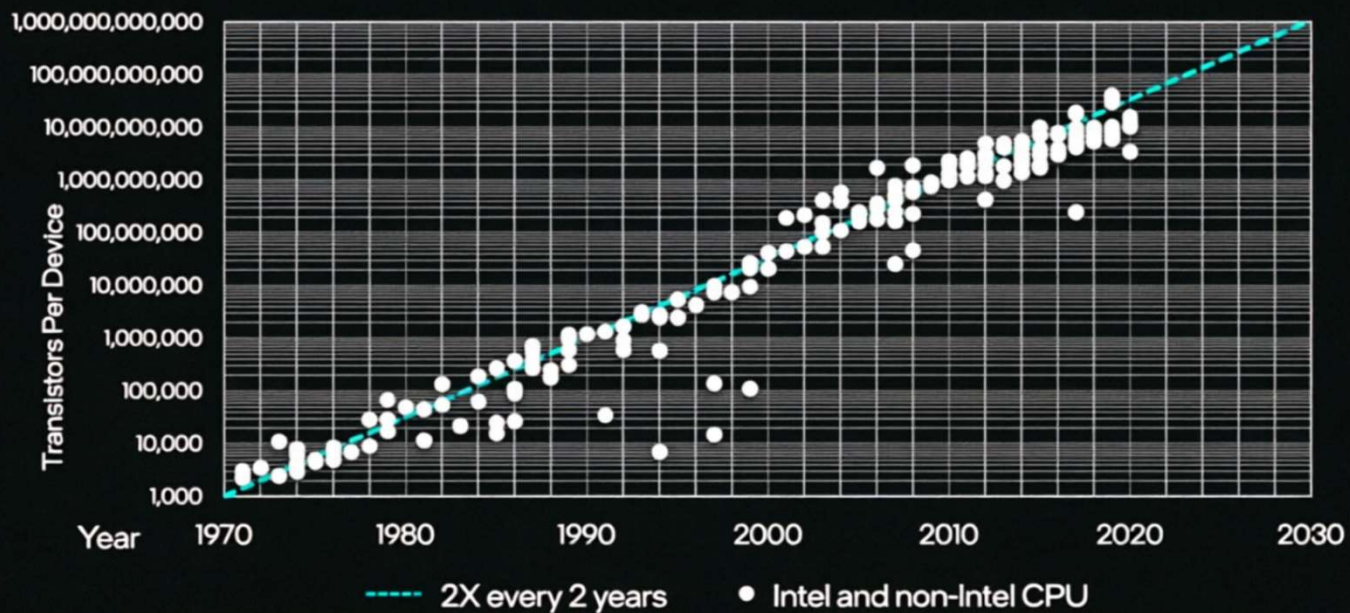
IC Manufacturing Trends

1965 Initial Prediction: Gordon Moore observed that chip complexity had been doubling annually since the invention of the integrated circuit and predicted it would continue for 10 years.

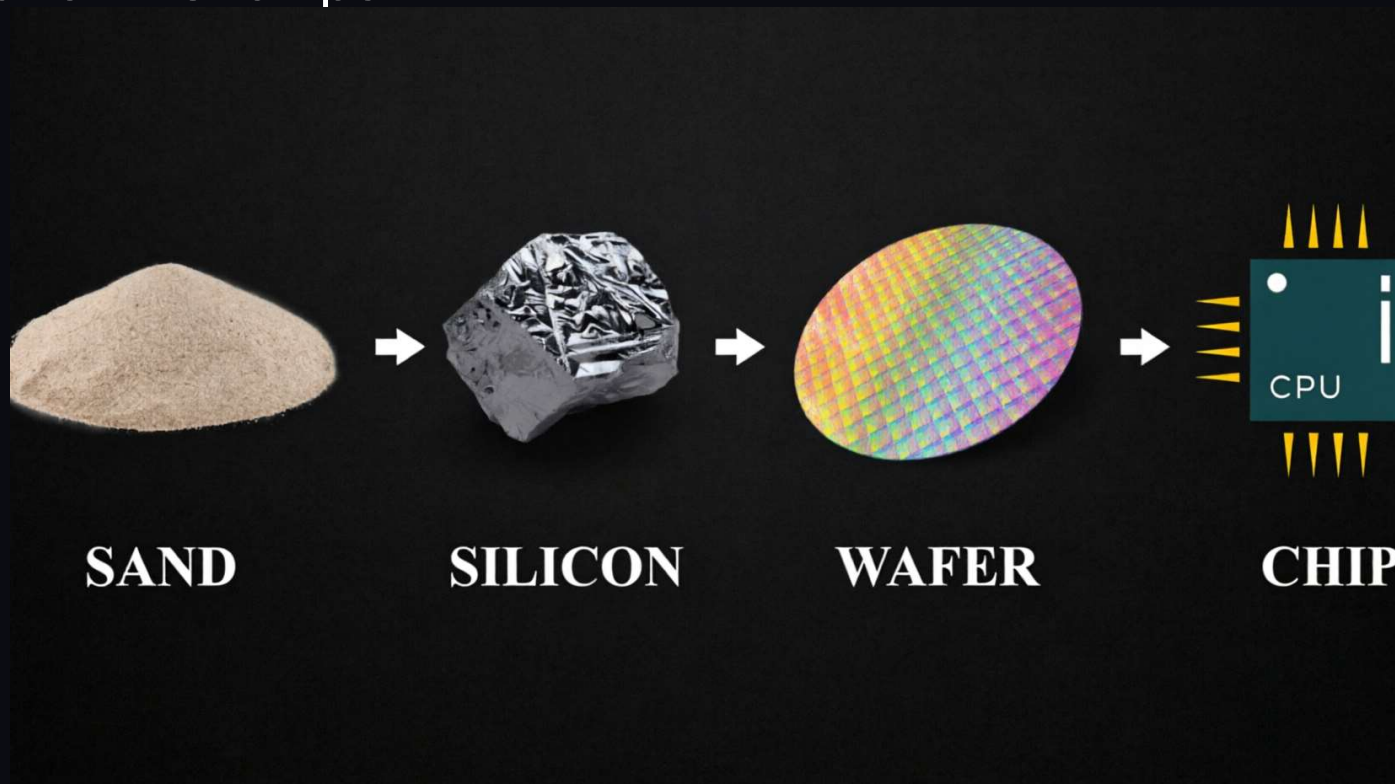
1975 Revision: Moore adjusted the forecast to a doubling of transistor count every two years to match actual manufacturing trends

Self-Fulfilling Prophecy: Tech companies adopted the observation as a synchronized manufacturing quota and performance target, driving intense multi-billion-dollar R&D cycles

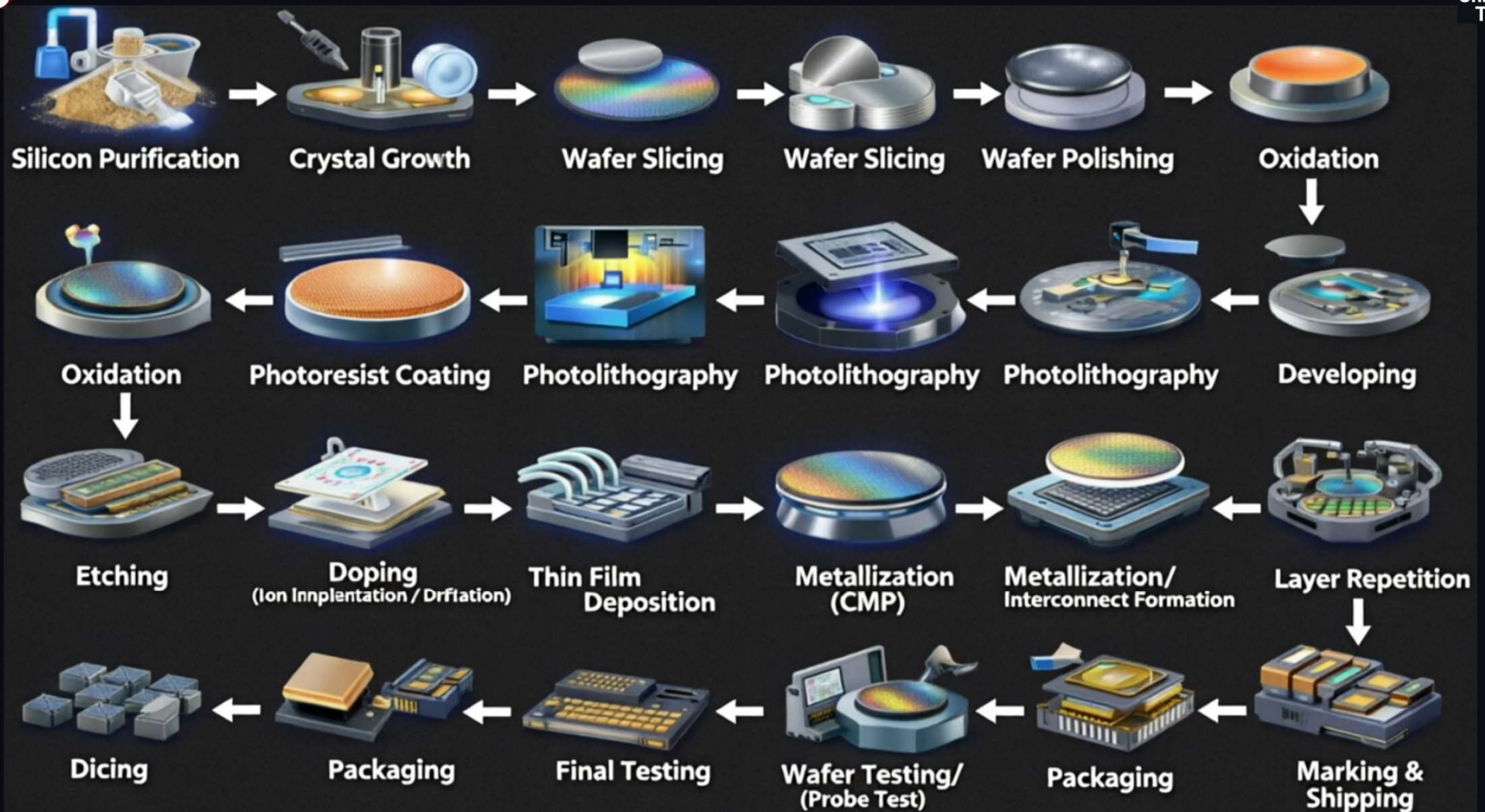
Moore's Law is alive and well



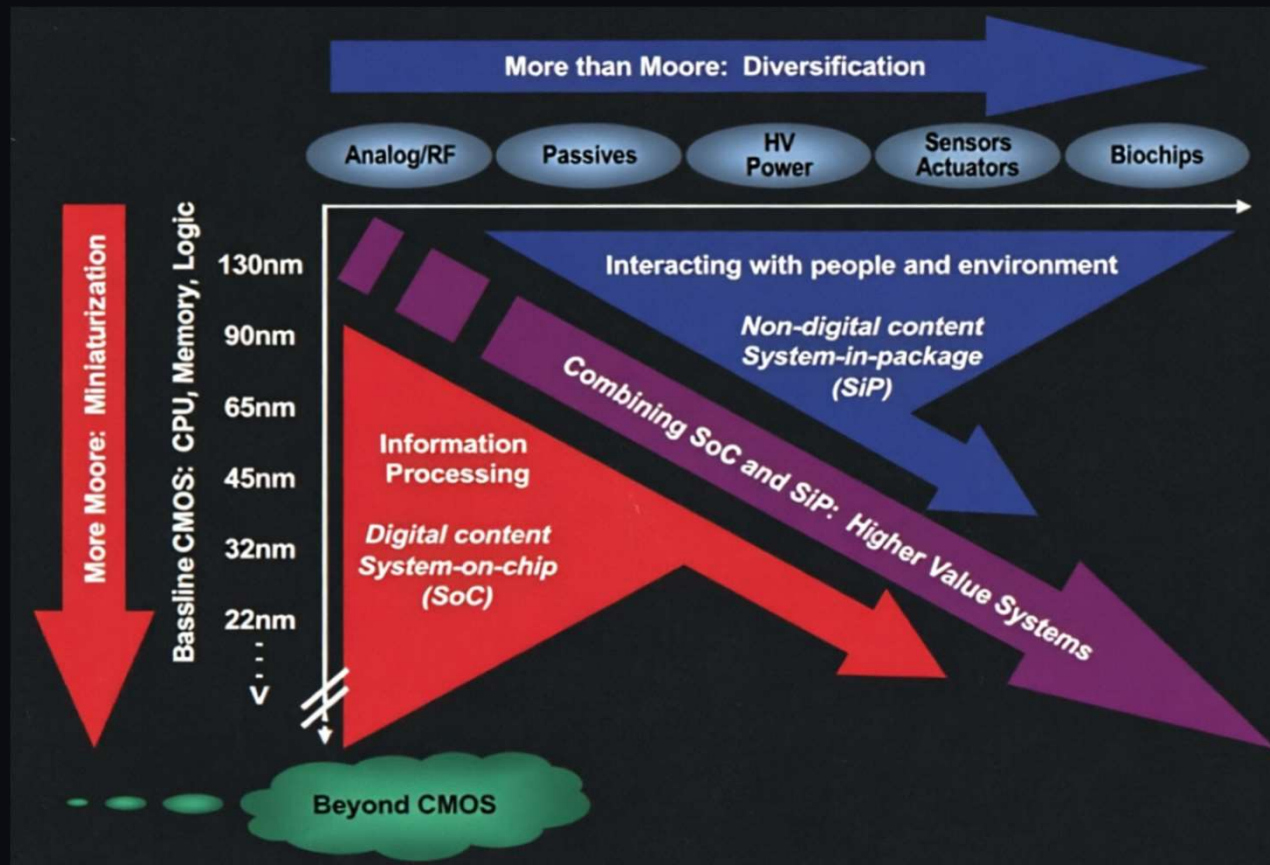
From Sand to VLSI chips



VLSI Manufacturing Process



More than Moore's Law



The end of Moore's Law & Scaling

Dennard scaling :

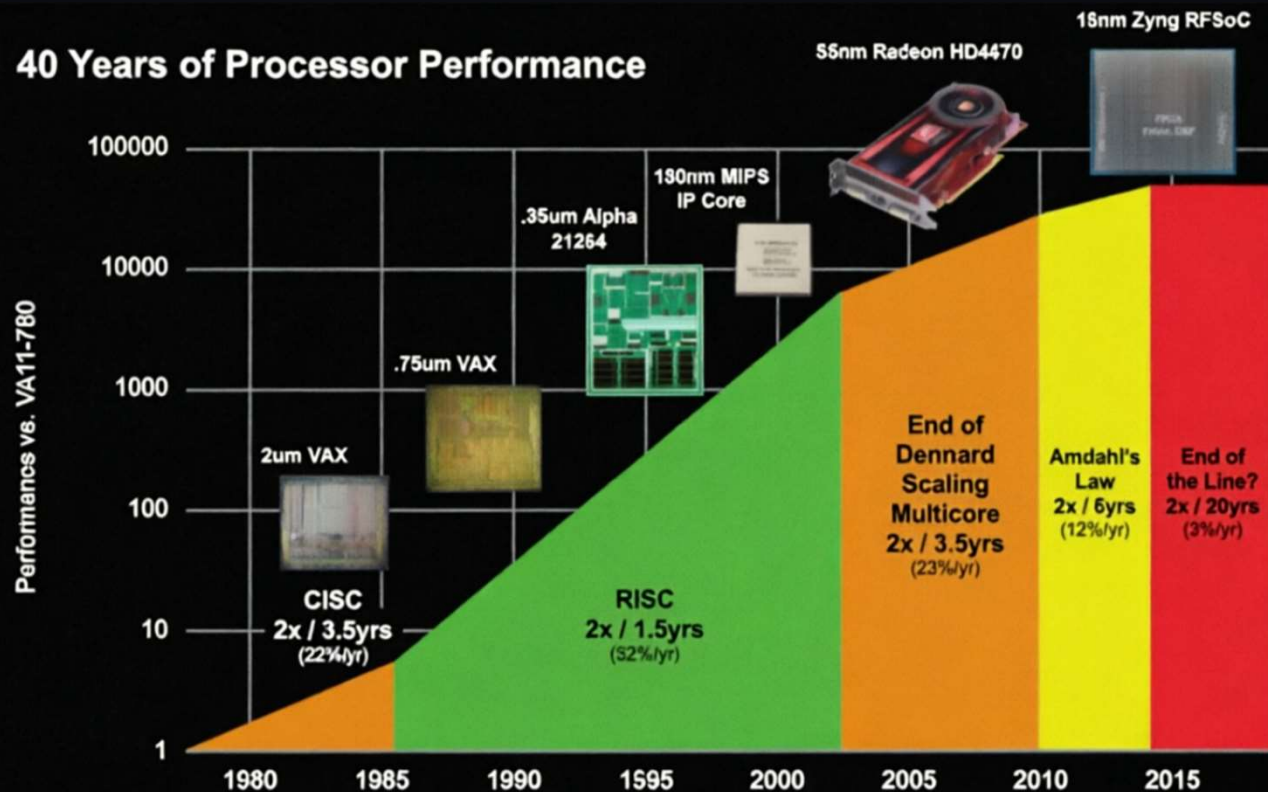
As transistors get smaller, their power density remains constant

Power consumption per unit area stays flat, meaning you the number of transistors on a chip could double (Moore's Law) and , run faster without increasing total power

Ahmdahl's Law :

Adding more hardware resources yields smaller and smaller performance gains over time

If 10% of a program cannot be parallelized, the maximum possible speedup is capped at 10×, regardless of core count



Source: John Hennessy and David Patterson, *Computer Architecture: A Quantitative Approach*, 6/e 2015

Types of Digital IC

By Integration Level

SSI (Small Scale Integration): 1–10 gates

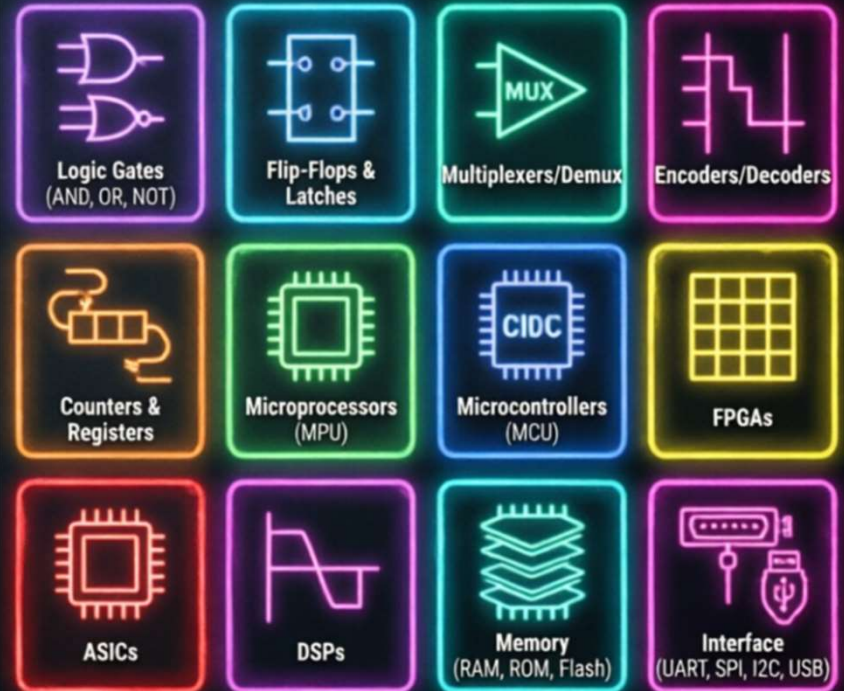
MSI (Medium Scale Integration): 10–100 gates

LSI (Large Scale Integration): 100–10,000 gates

VLSI (Very Large Scale Integration):
10,000–1,000,000 gates

ULSI (Ultra Large Scale Integration):
1,000,000+ gates

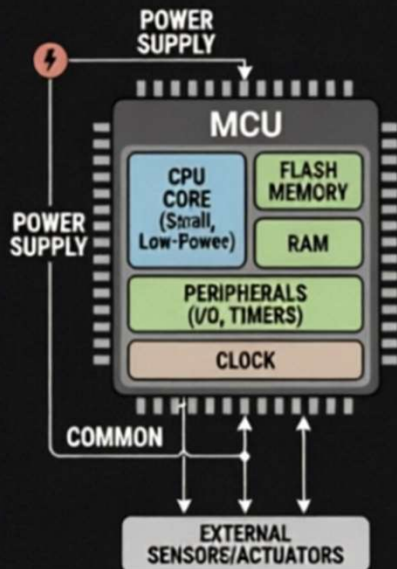
By Function



FUNCTIONAL PHYSICAL ARCHITECTURE DIFFERENCES: MCU, MPU, SOC, FRGA

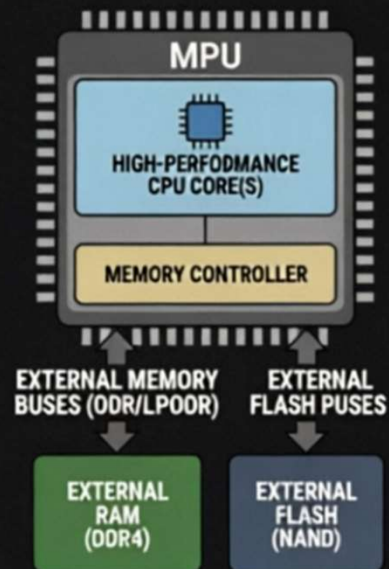
MCU

(Microcontroller Unit)
All-in-One, Low-Speed



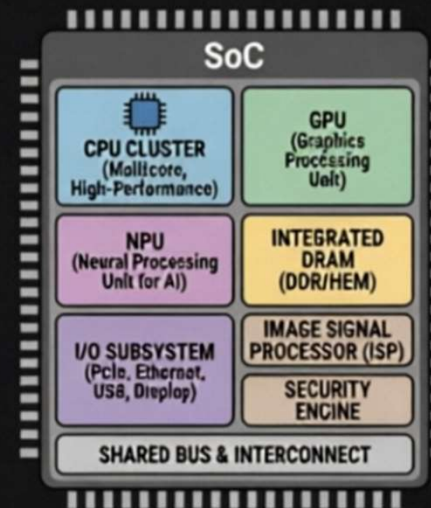
MPU

(Microprocessor Unit)
Raw CPU + External Memory



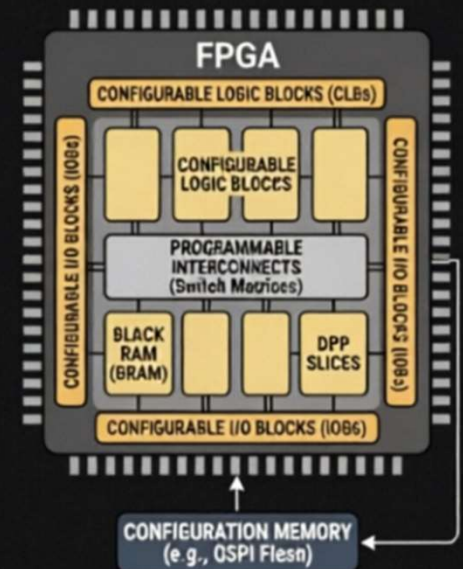
SoC

(System-on-Chip)
Highly Integrated, High-Speed



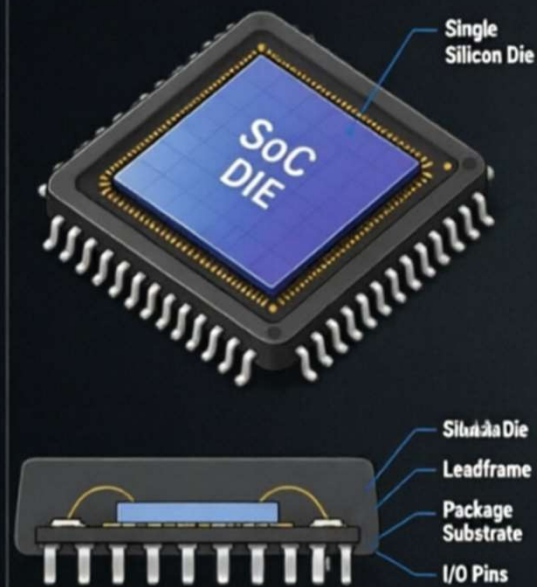
FPGA

(Field-Programmable Gate Array)
Custom Programmable Logic Array

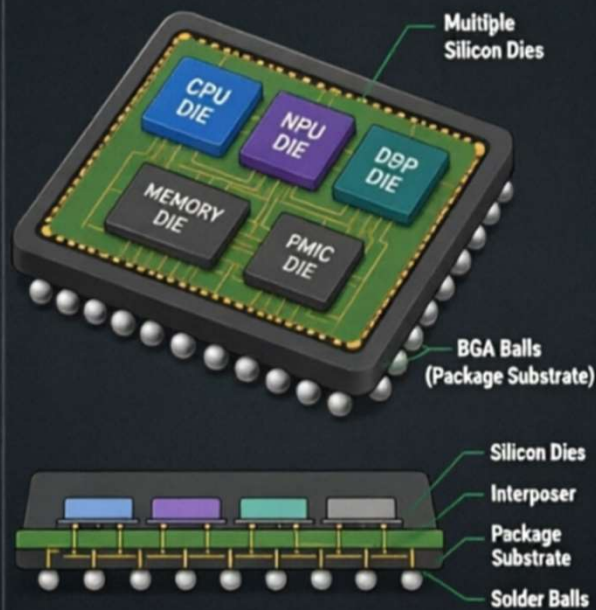


System Integration Level

1. SoC System on Chip



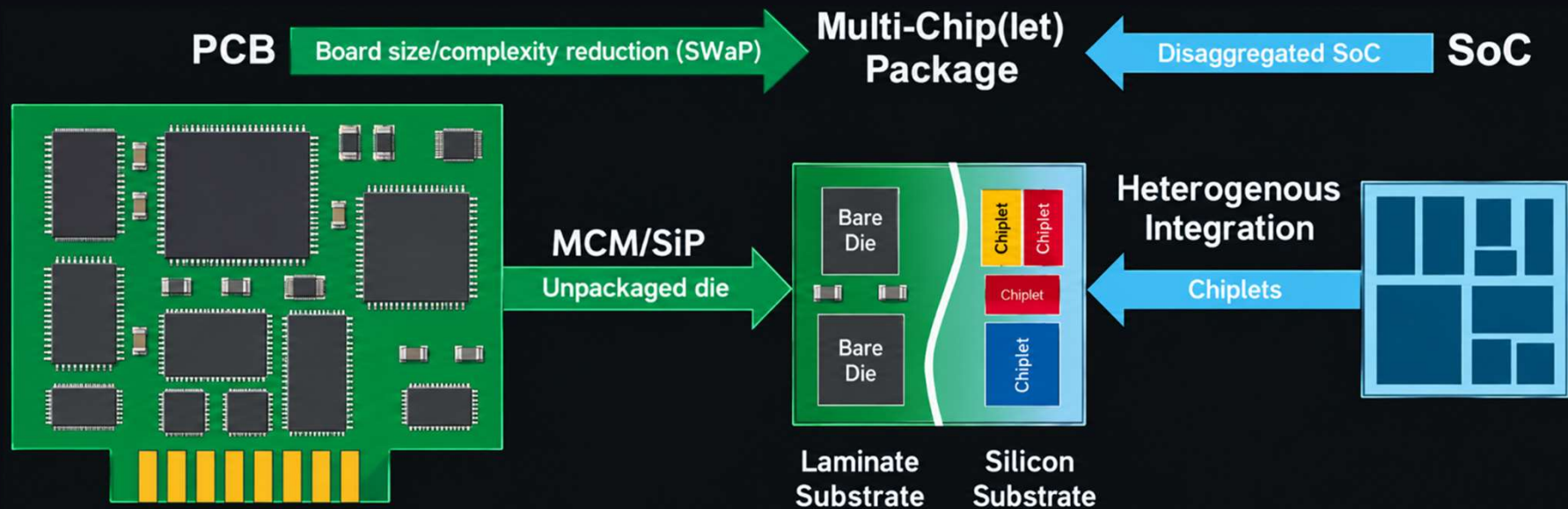
2. SiP (BGA) Multiple Dies in One Package



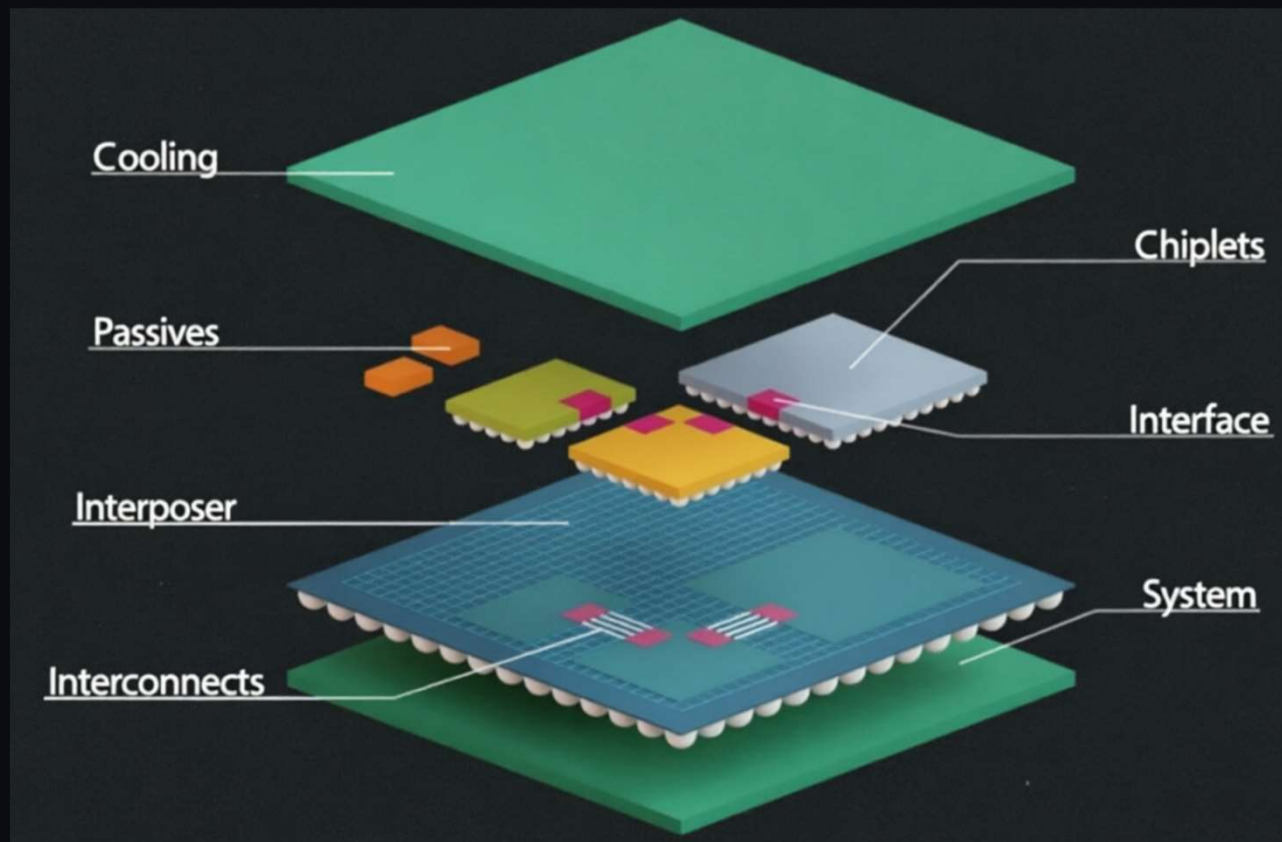
3. SoM (System on Module) Pre Built Module (Bcard Level)



Chiplet : Multi Chip Integration

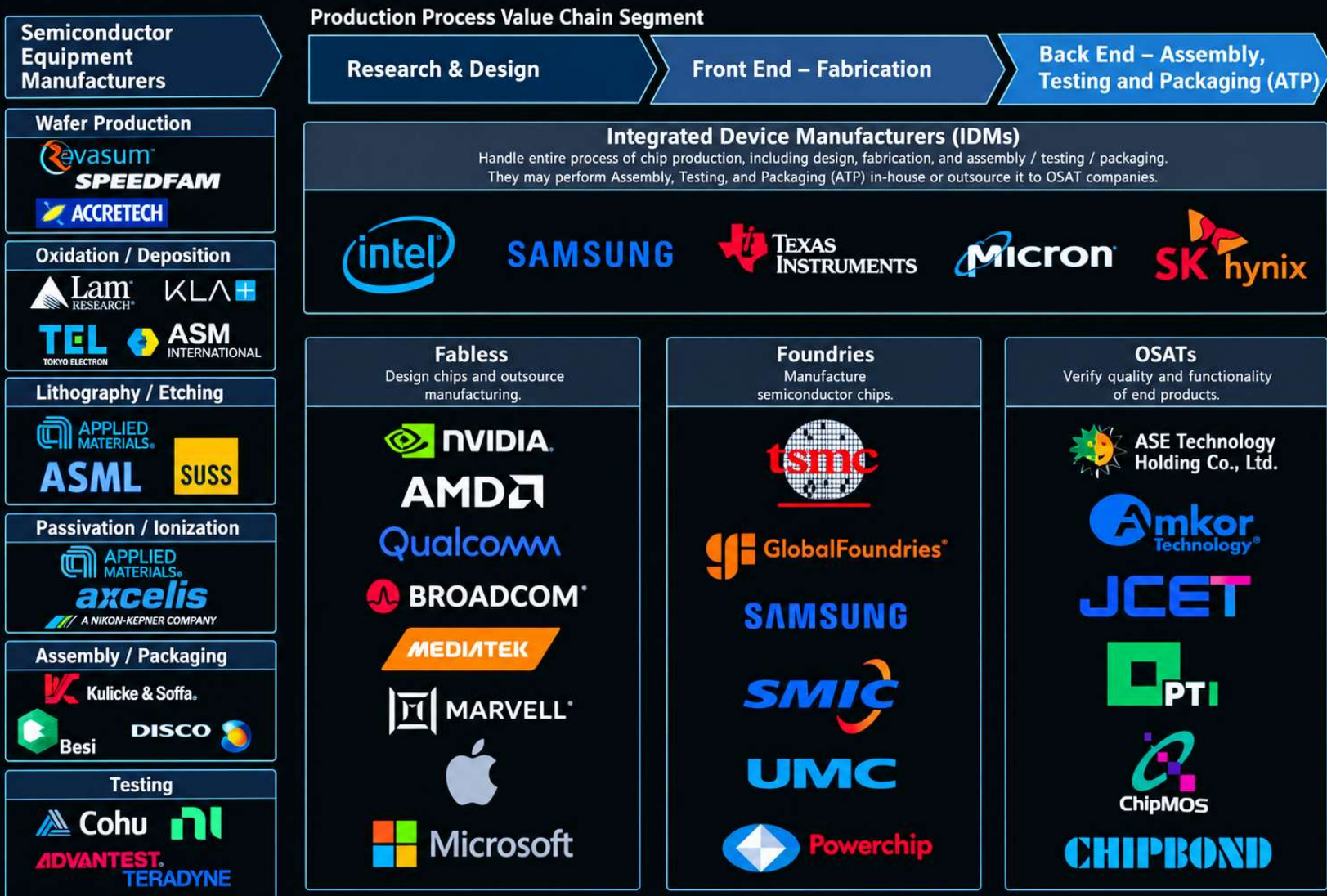


Structure of Chiplet





VLSI Ecosystem



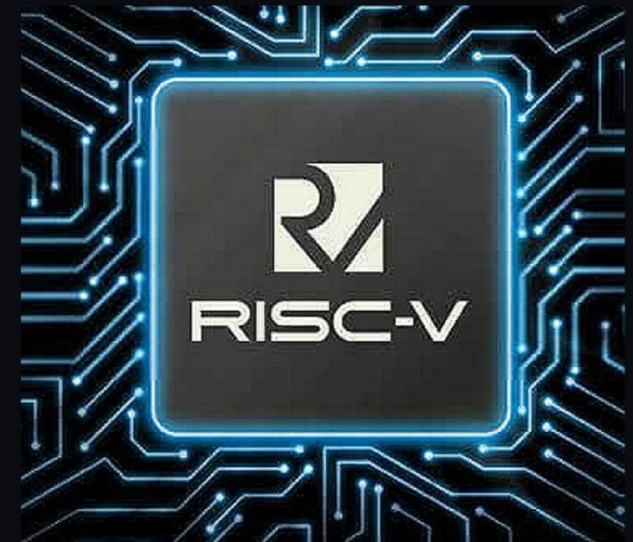


RISC-V : Open (standard) Processor

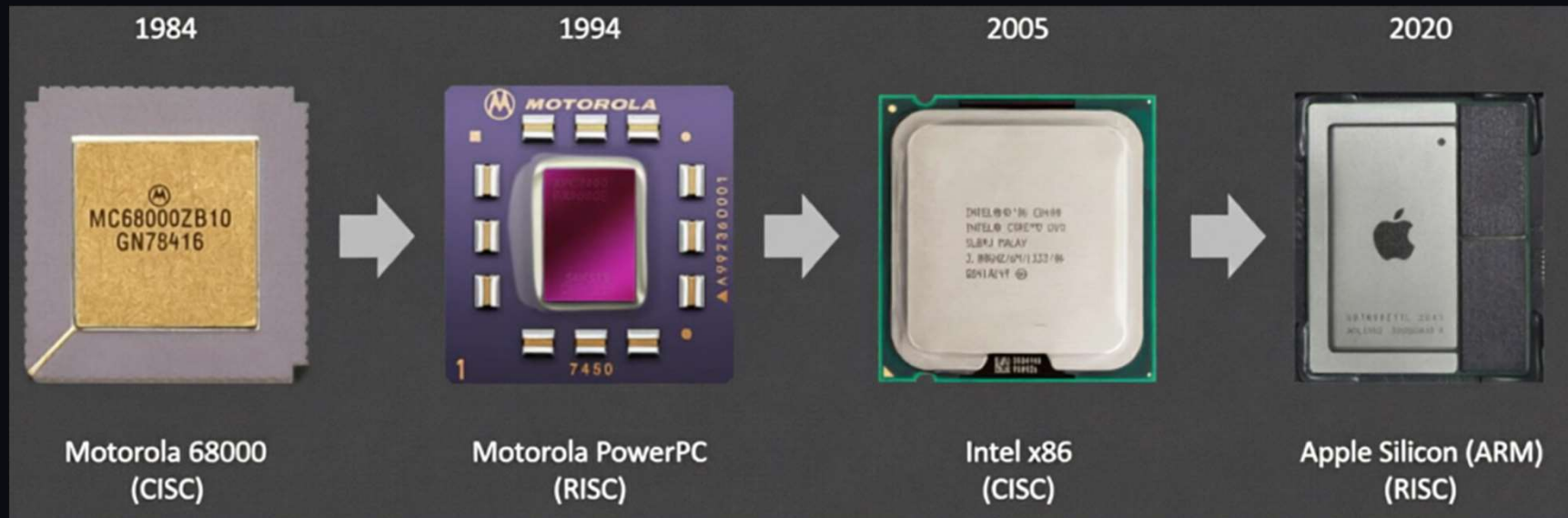


What is RISC-V

- **Open Standard:** Unlike proprietary architectures like x86 (Intel/AMD) or ARM, RISC-V specifications are free to use.
- **No Royalties:** Anyone can design, manufacture, and sell RISC-V chips without paying expensive licensing fees.
- **Modular Design:** It uses a simple base instruction set with optional extensions, making it flexible for everything from tiny microcontrollers to massive data center servers
- **Origin (UC Berkeley):** The project started in 2010 by researchers led by Professor Krste Asanović at the University of California, Berkeley to create a free hardware design



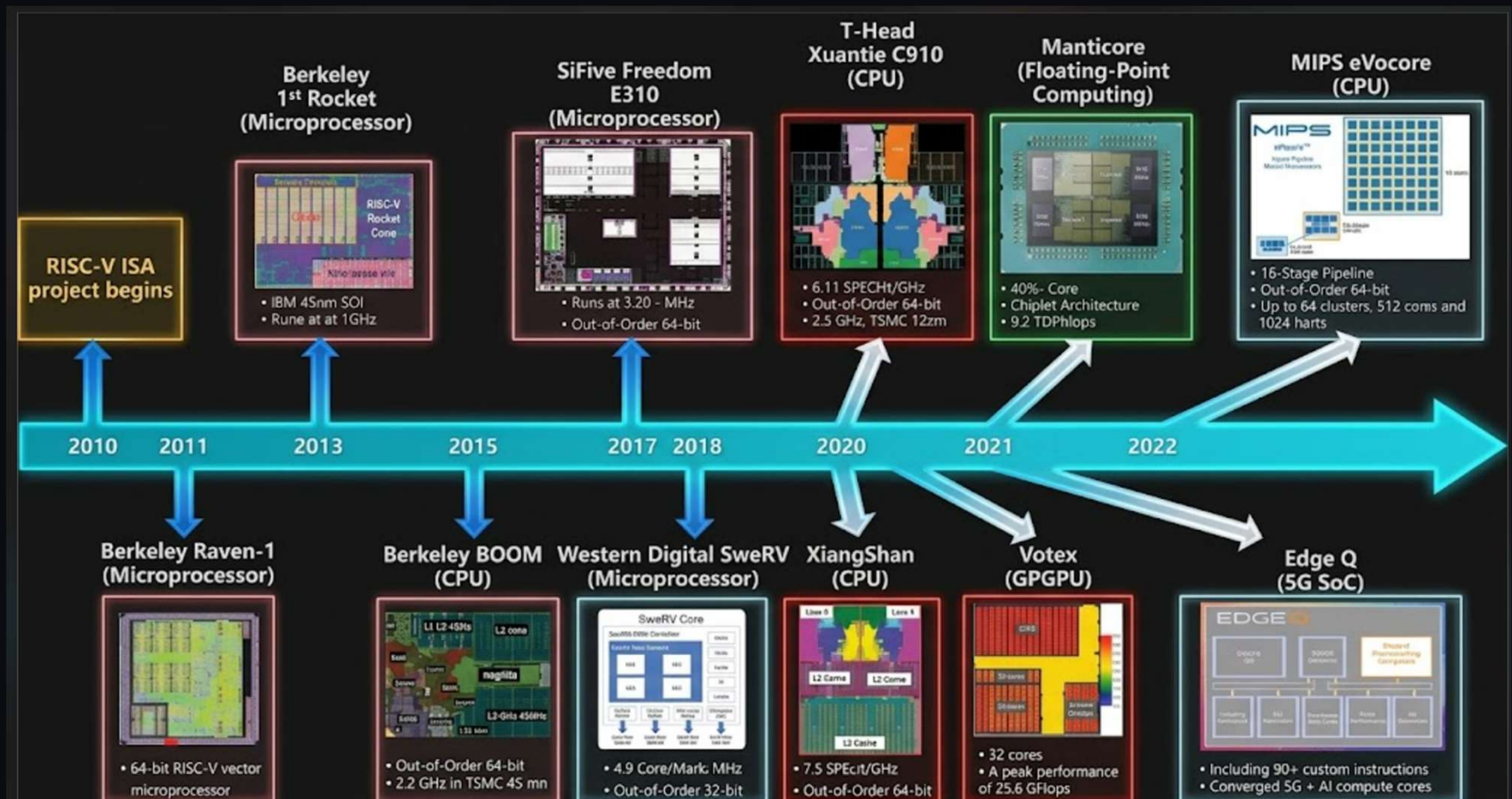
RISC : Low Power Processor



Complex Instruction Set Computer (CISC) Vs Reduced Instruction Set Computer (RISC)

(Apple Macintosh CPU transition timeline)

Evolution of RISC-V





RISC-V Foundation Members

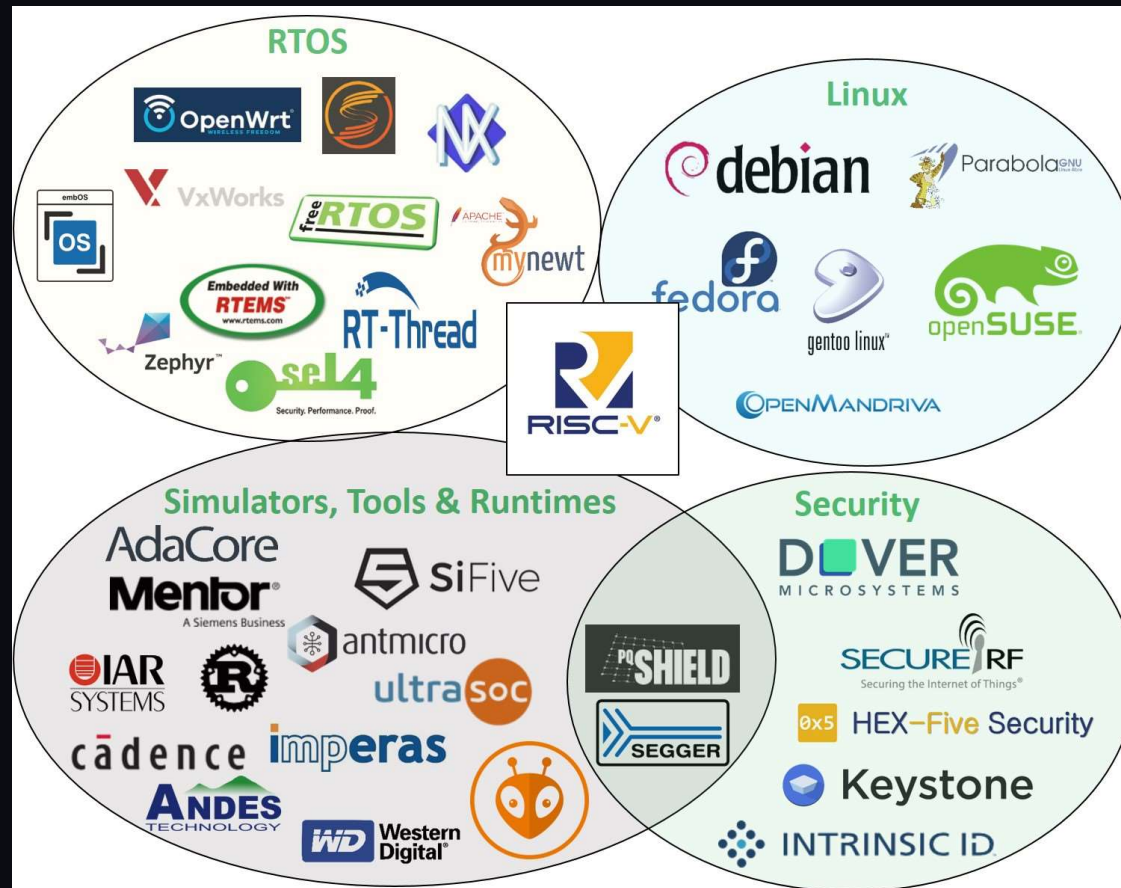


RISC-V Foundation: 65+ Members





RISC-V Ecosystem

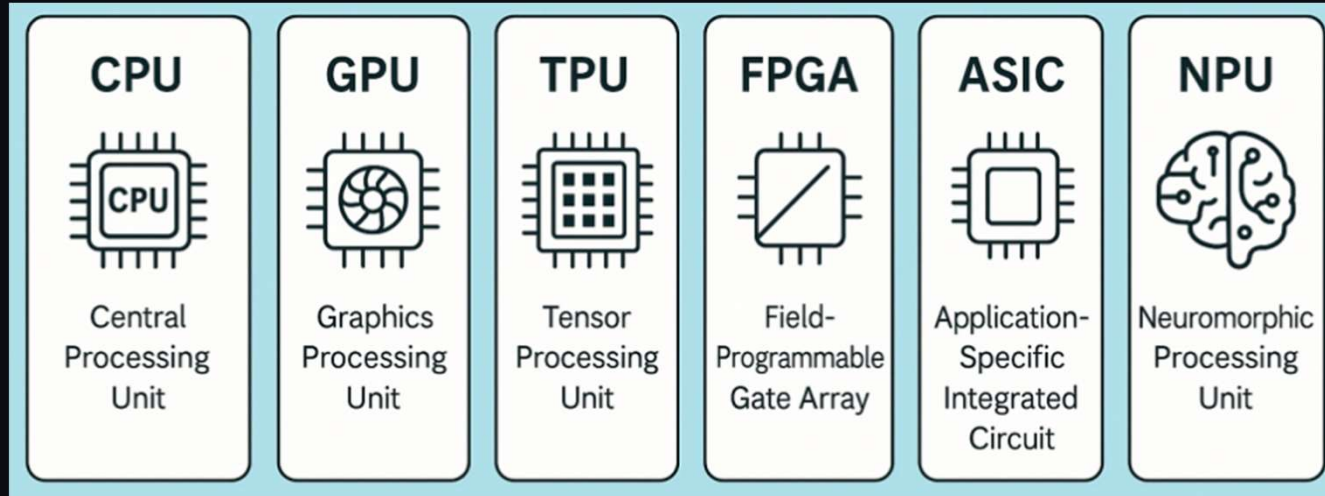




VLSI for Artificial Intelligence

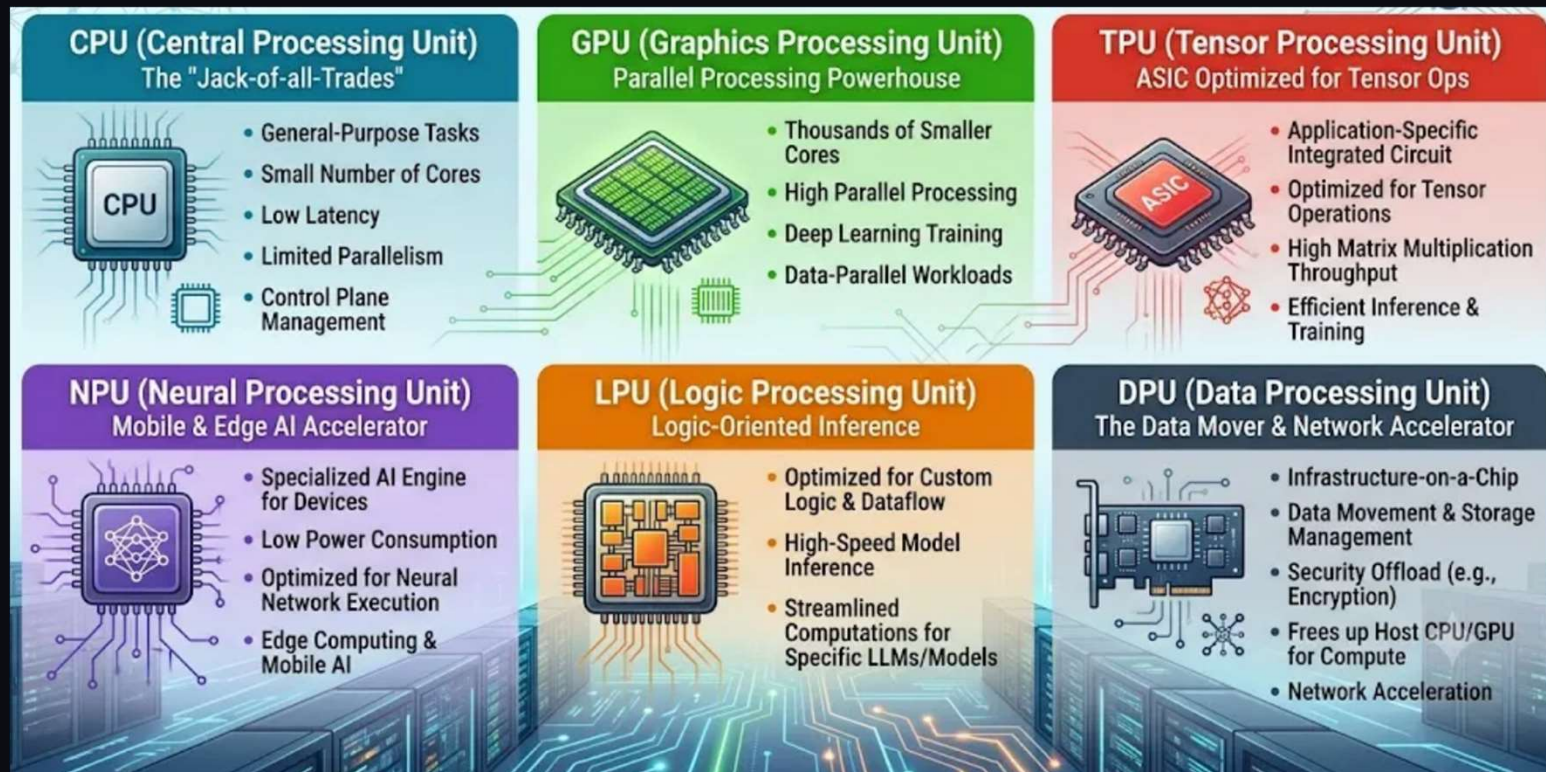


AI Engine

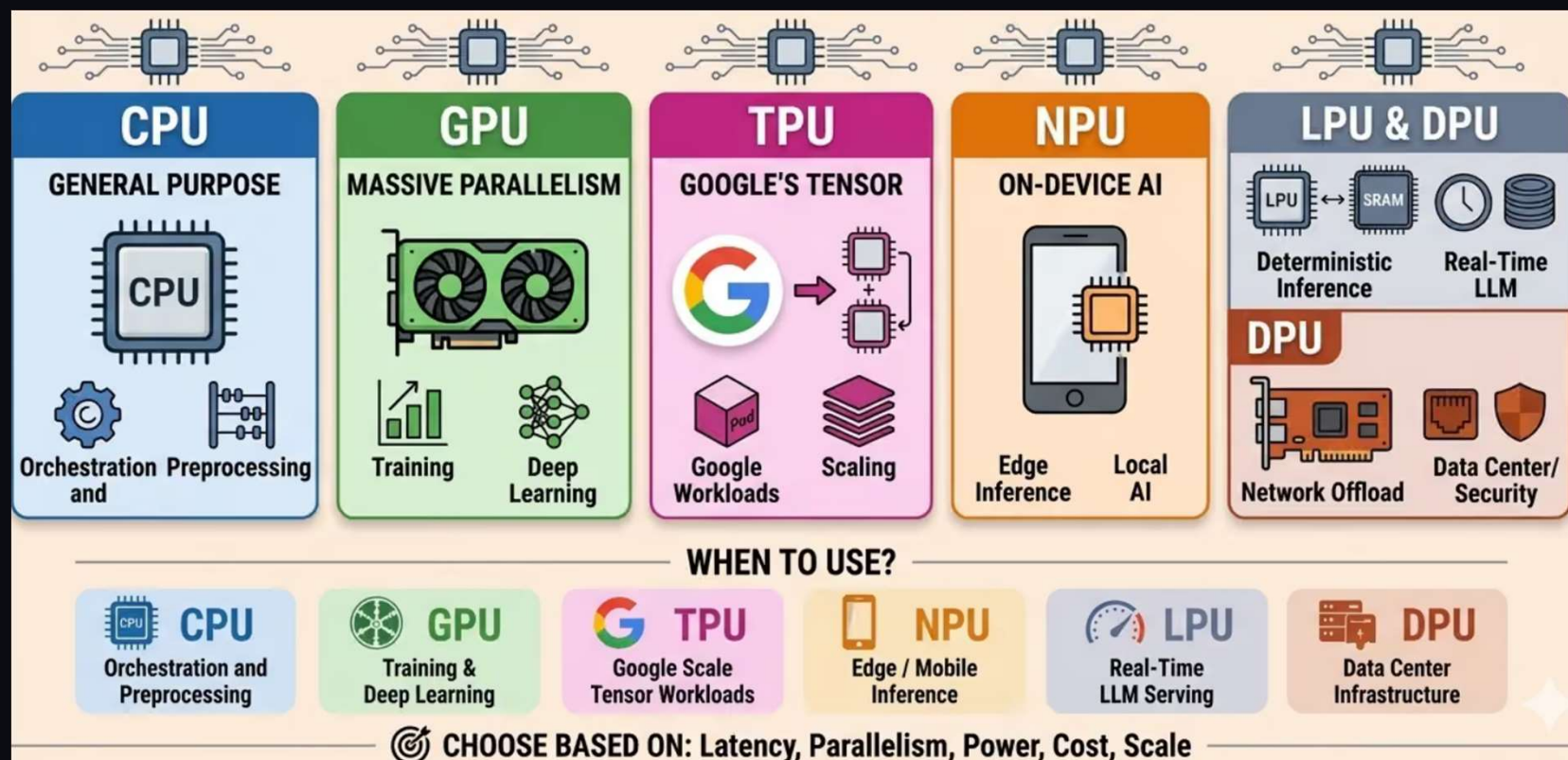


- Central Processing Units : The All-Purpose Thinkers
- Graphics Processing Units : Masters of Parallel Thinking
- Tensor Processing Units : AI-Specific Accelerators
- Field-Programmable Gate Arrays : The Customizable Problem-Solvers
- Application-Specific Integrated Circuits: The Dedicated Specialists
- Neuromorphic Processing Units : Mimicking the Brain

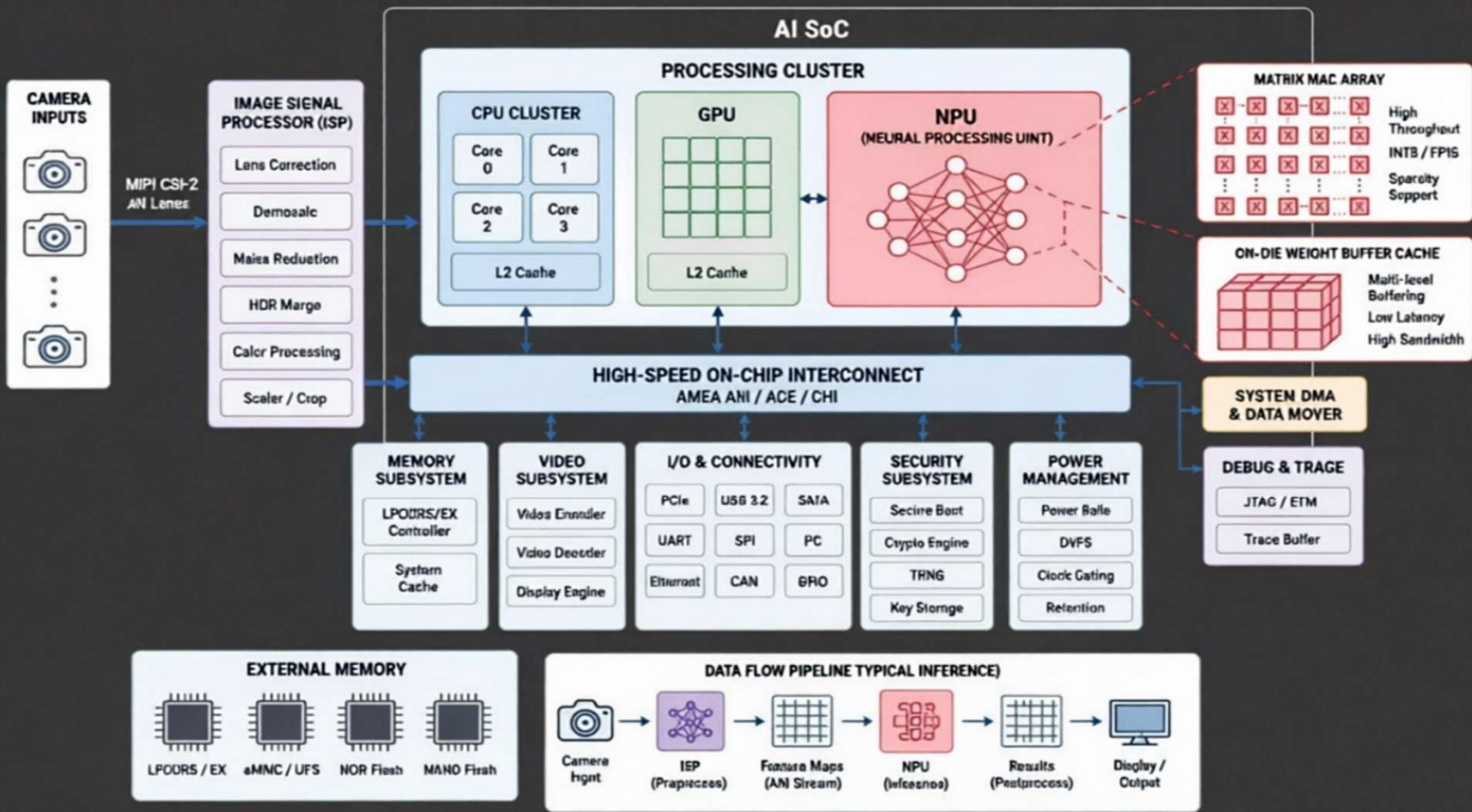
Different AI Device for Different Purpose



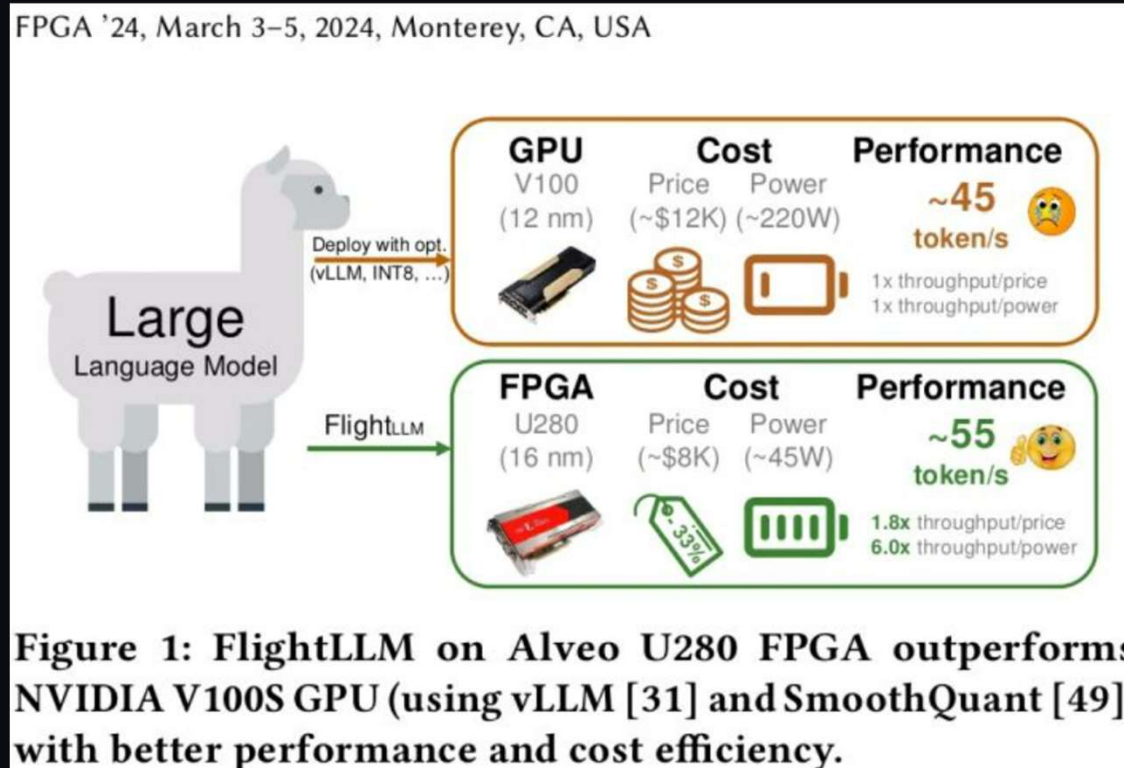
Choosing AI Device



AI SoC SYSTEM ARCHITECTURE



Costum Chip (FPGA) vs GPU for LLM





Lets Start Design - VLSI



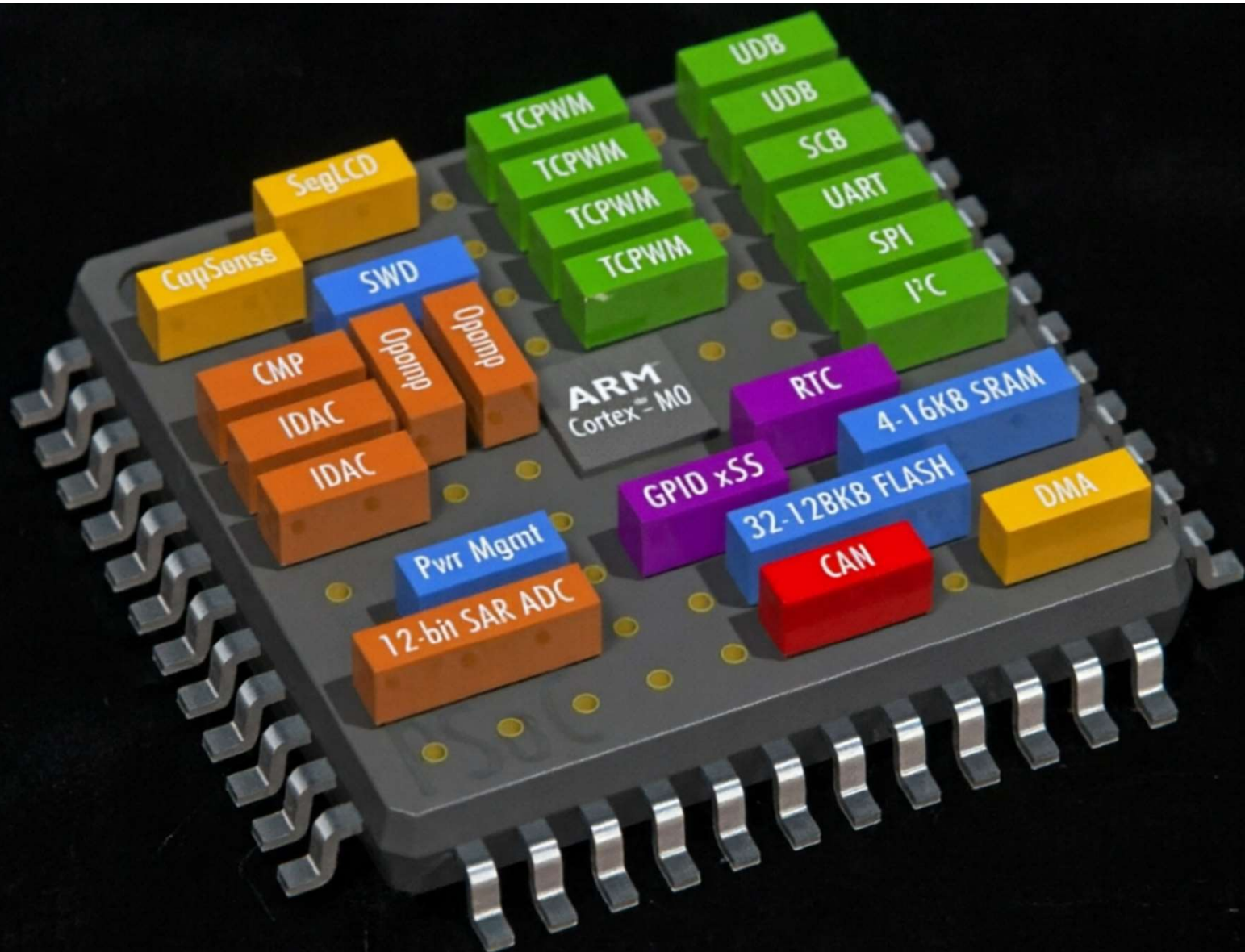


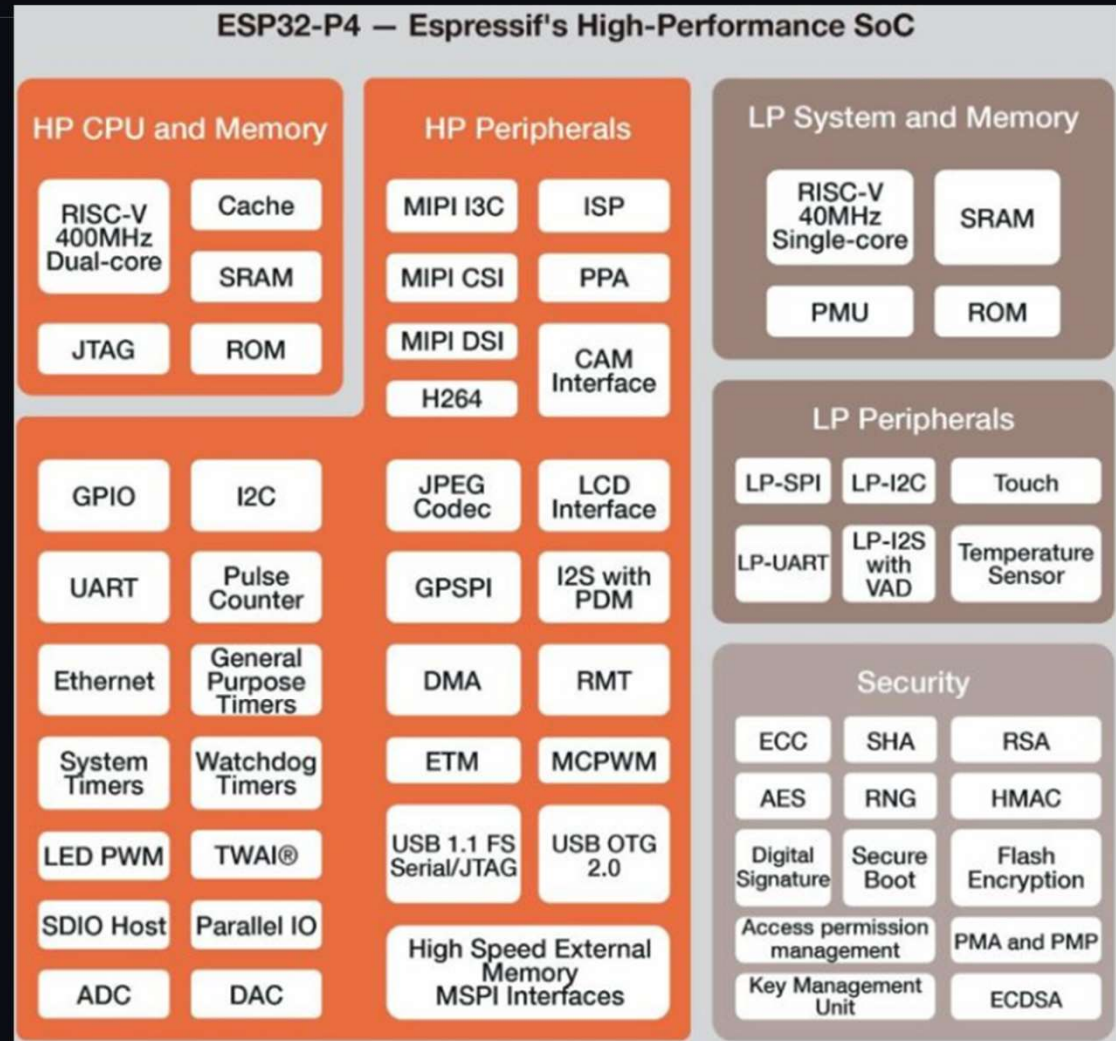
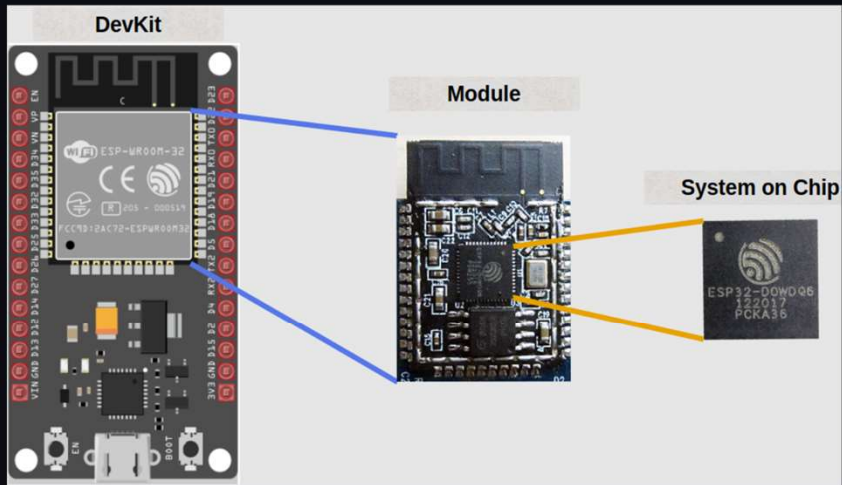
Thank You

Nandri (நன்றி)

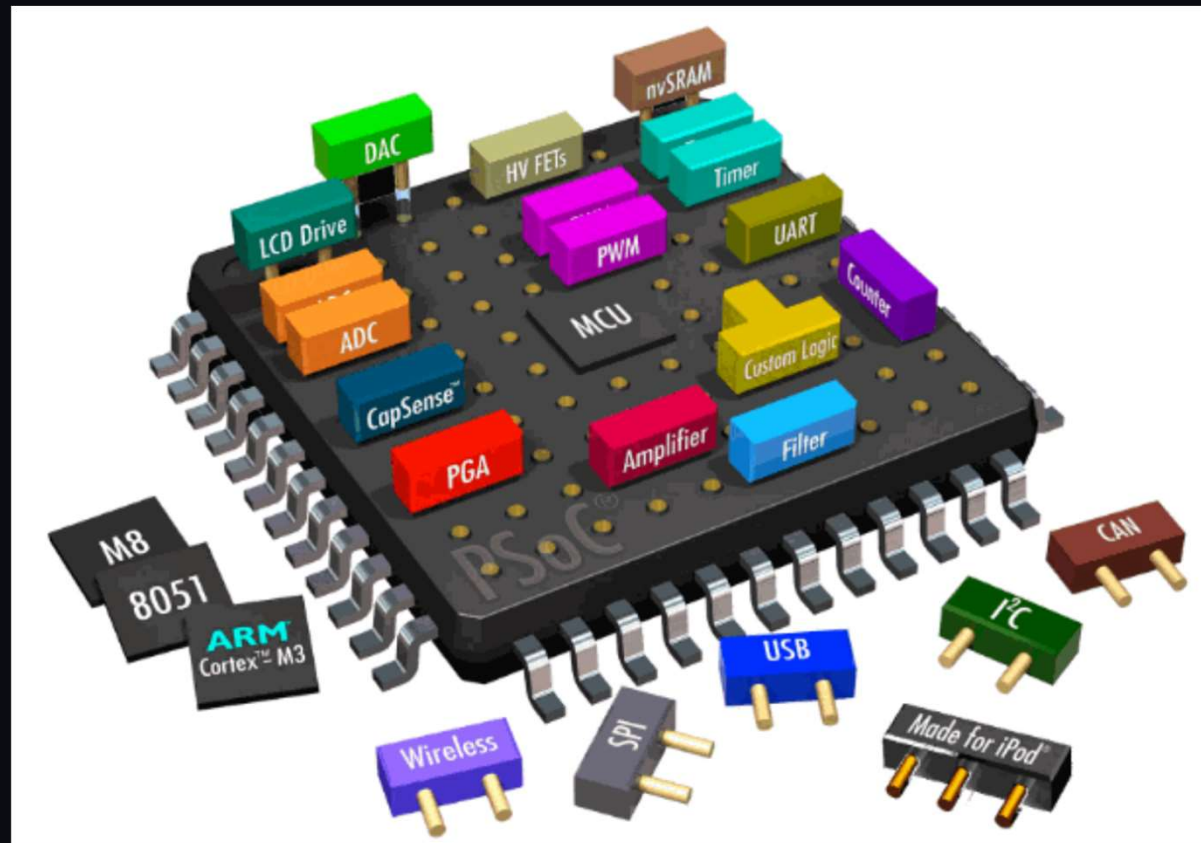
Terima Kasih

System on Chip

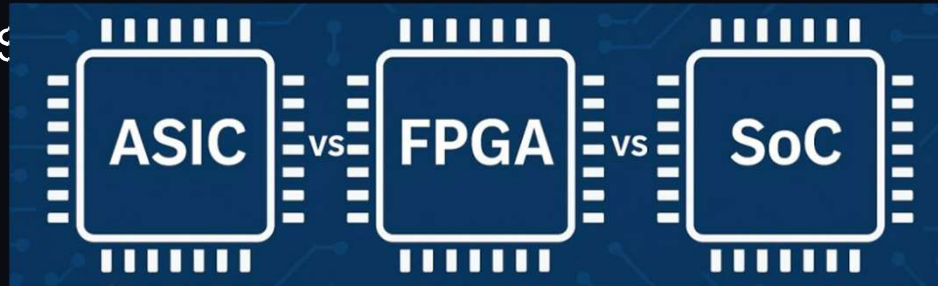


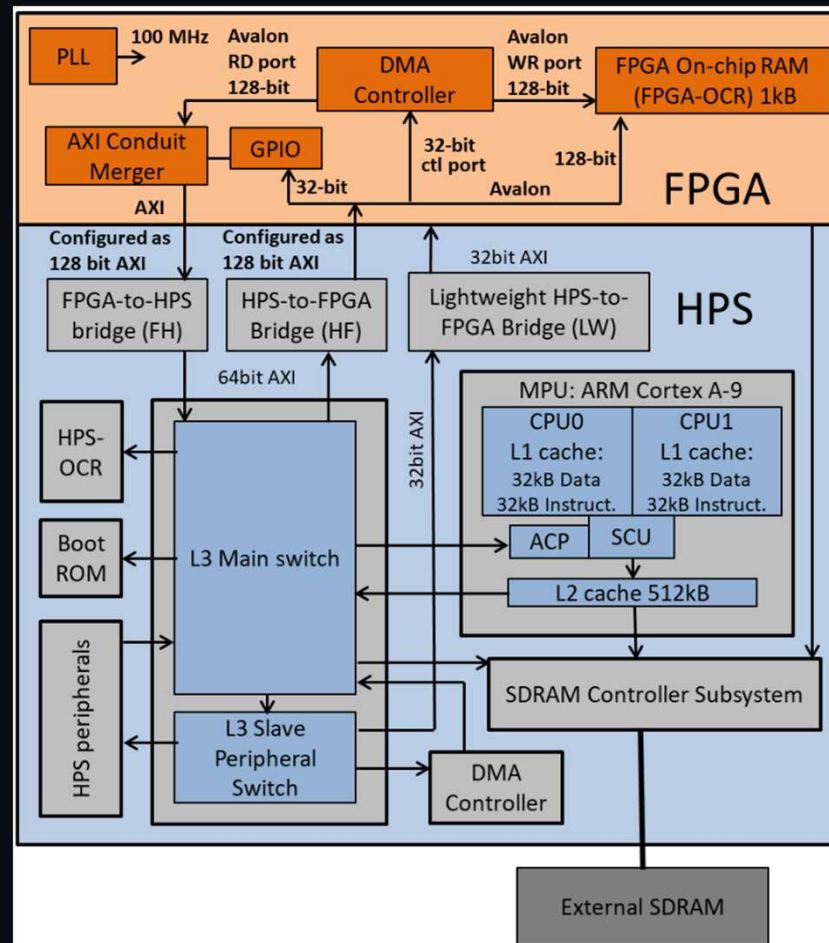


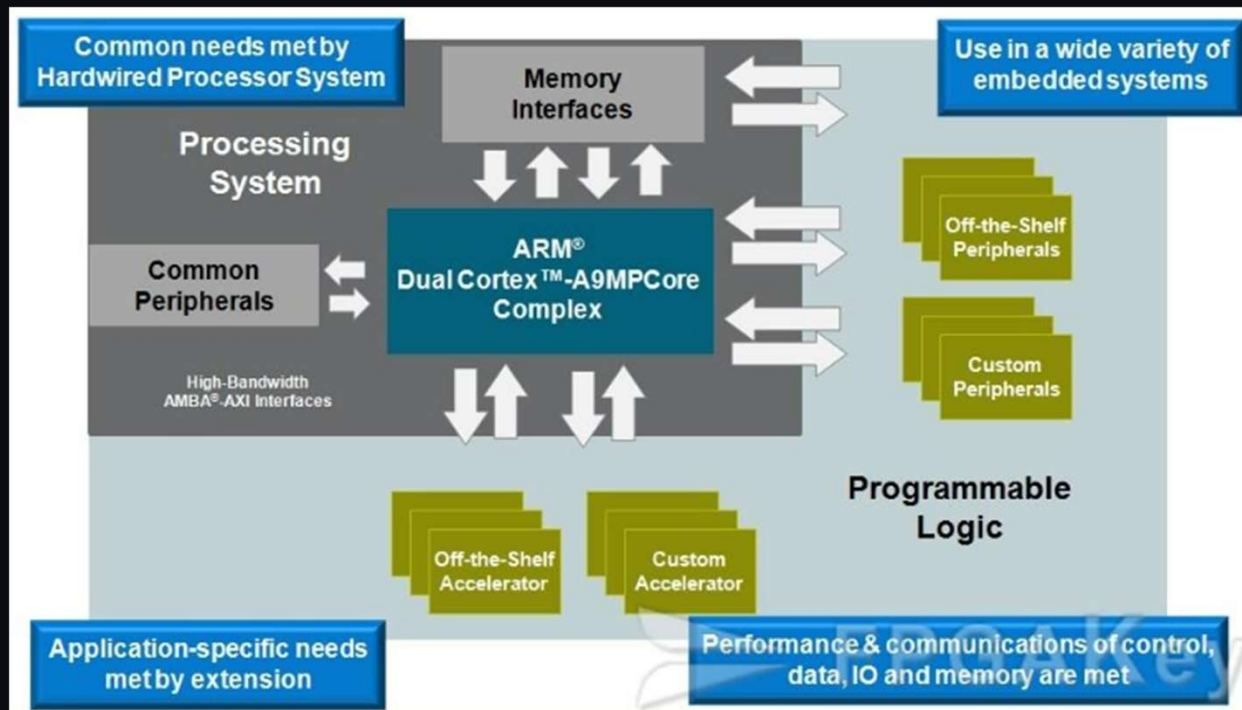
FPGA-SoC

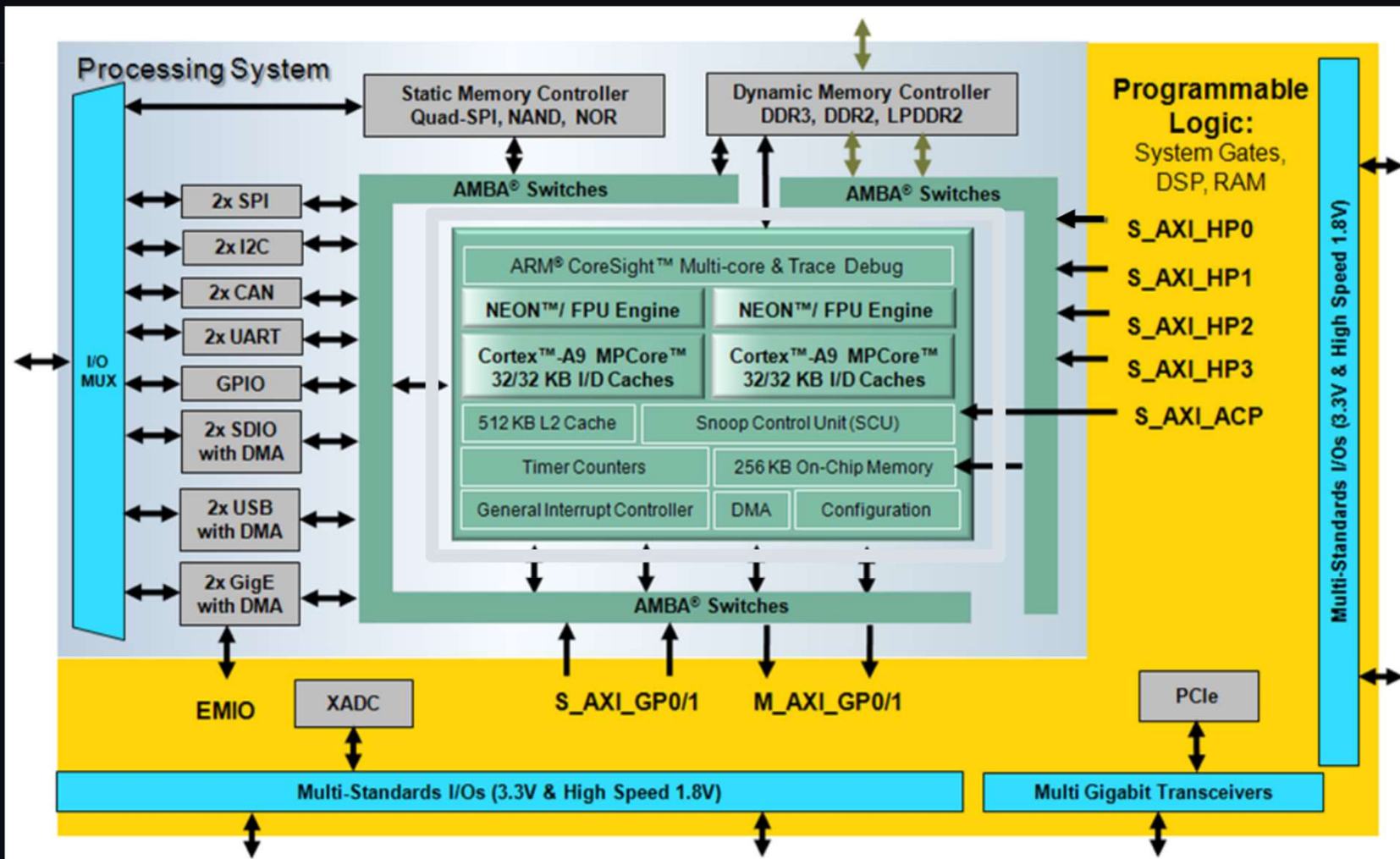


ASIC vs FPGA vs









FPGA in Modern Application



Real-Time Processing



Parallel Pipeline



Safety & Reliability

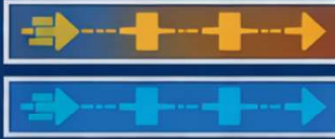
Sensors



Perception Inputs

Preprocessing
ISP & Filtering

FPGA-Based
Physical AI
Architecture



Decision &
Control Logic

Actuators



Real-World Actions

Sensor Fusion

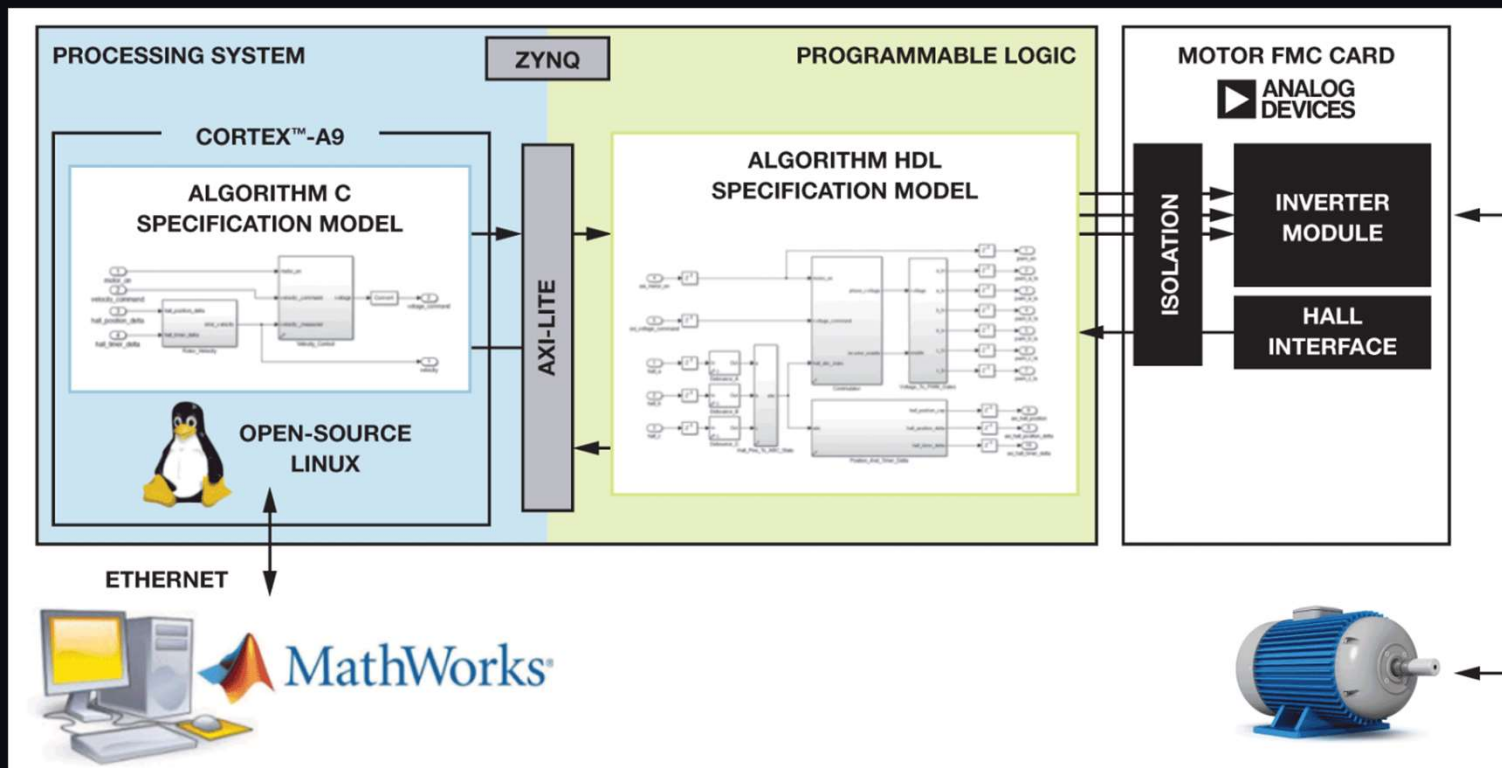
Edge AI Acceleration

Hard Real-Time Control

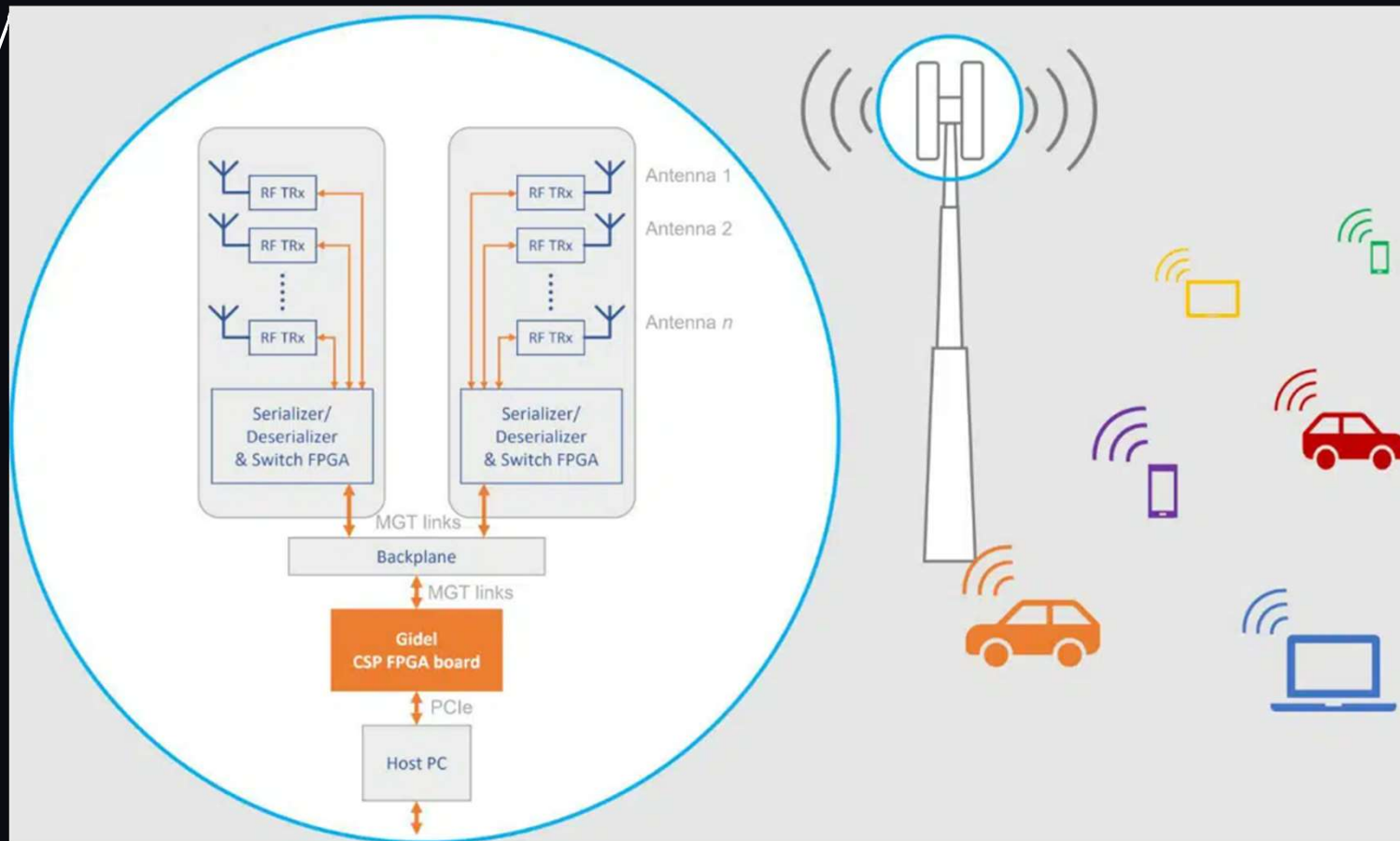
Safety Isolation



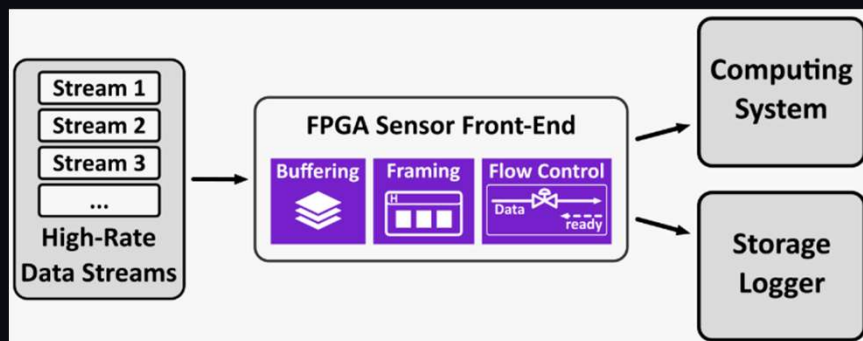
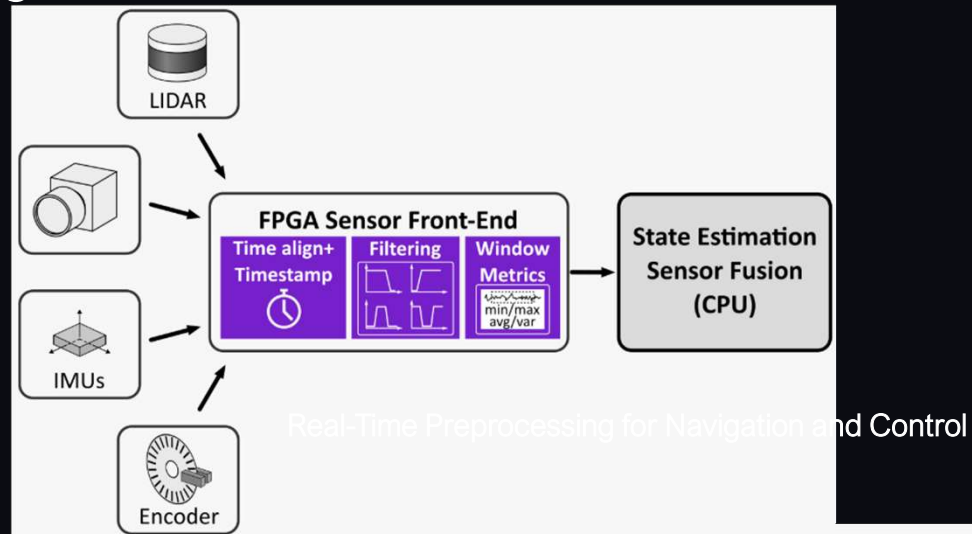
Tesla's first-generation Autopilot hardware (Hardware 1.0) did indeed rely on [FPGAs \(Field-Programmable Gate Arrays\)](#) as a critical component of its processing architecture.



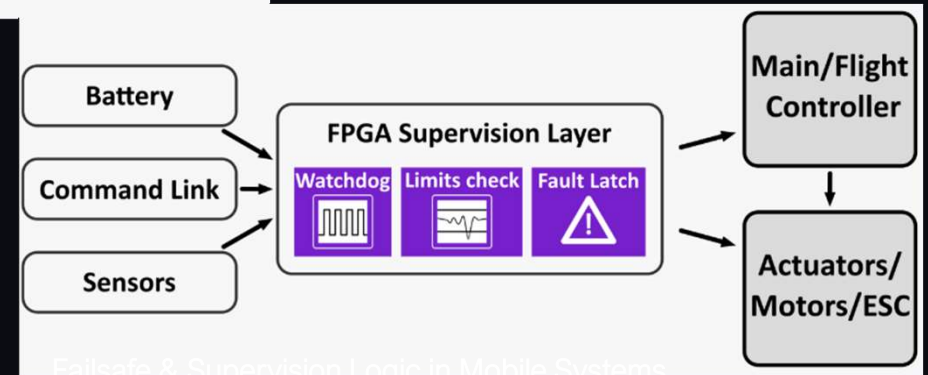
Massive M



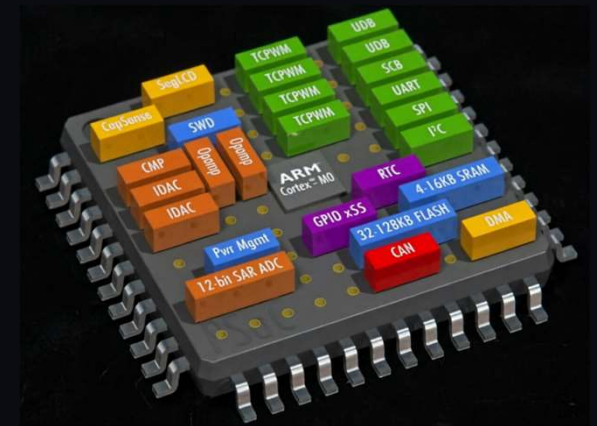
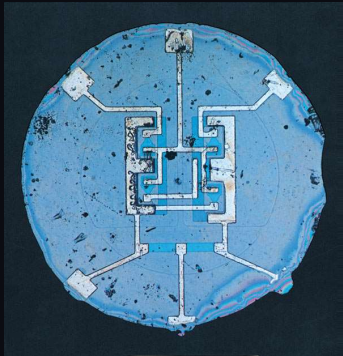
Robotics and UAVs



High-Rate Data Handling for Edge Perception



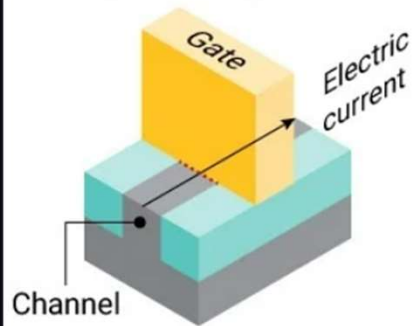
Fail-safe & Supervision Logic in Mobile Systems



Generations of transistors

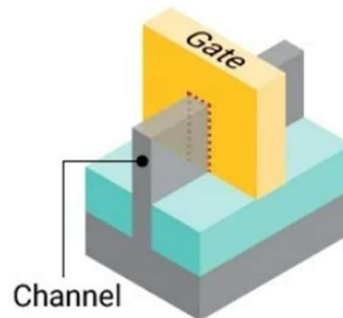
Gates allow and prevent the flow of electric current, switching transistors on and off. They enable chips to receive, send and process digital data.

The greater the surface area between gate and channel (denoted by dotted red lines), the better the control and smaller the leakage of electric current, leading to a more powerful and efficient chip.



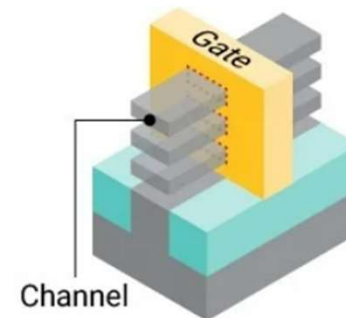
Planar

In use since 1960s
Size: 22 nm and larger
Classic transistor used in everyday electronics, such as refrigerators, microwaves, cars



FinFET*

In use since 2010s
Size: 22-3 nm
Used in many modern electronics; Apple's A17 Pro processor chip features 19 billion of these

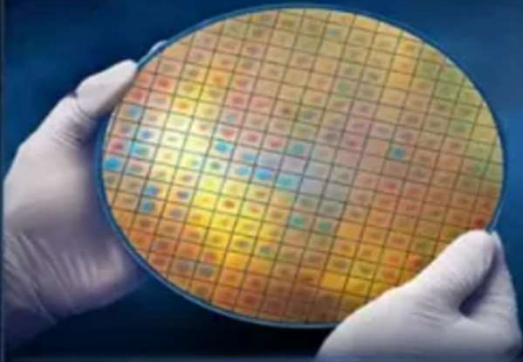


Gate-all-around (GAA)

In use since 2022**
Size: 3 nm and smaller
For high-performance computing systems like AI servers, autonomous driving technology

Semiconductors in Modern Electronics

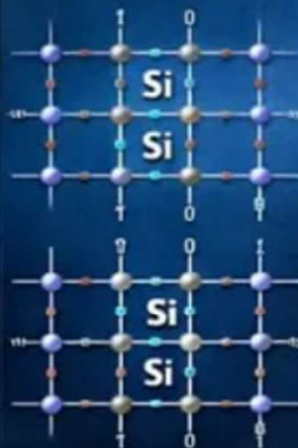
What is Semiconductor?



Material that has conductivity between a conductor and an insulator.

Types of Semiconductors

Intrinsic



Extrinsic

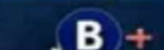
N-Type

+ Donor



P-Type

- Acceptor



Semiconductor Materials



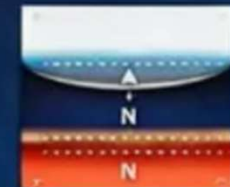
Silicon (Si)



Gallium Arsenide (GaAs)



Semiconductor Physics



Conduction Band

Forbidden Gap



Electron

Hole

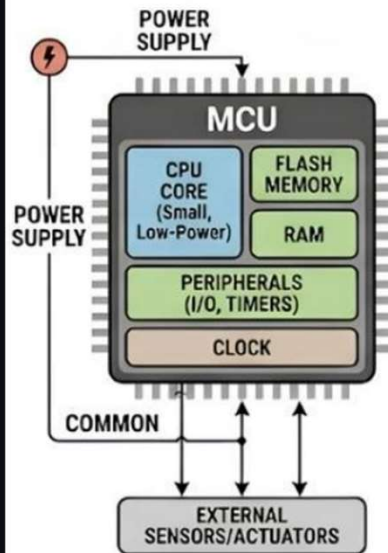
Depletion Region



FUNCTIONAL PHYSICAL ARCHITECTURE DIFFERENCES: MCU, MPU, SOC, FPGA

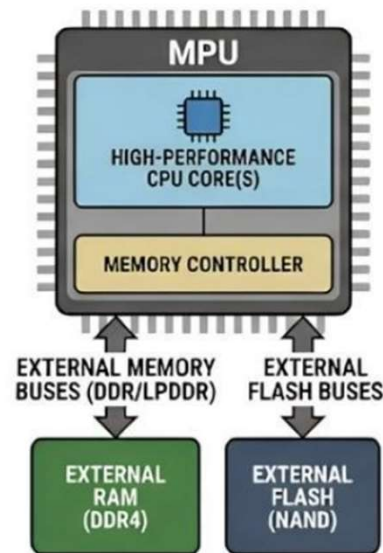
MCU

(Microcontroller Unit)
All-in-One, Low-Speed



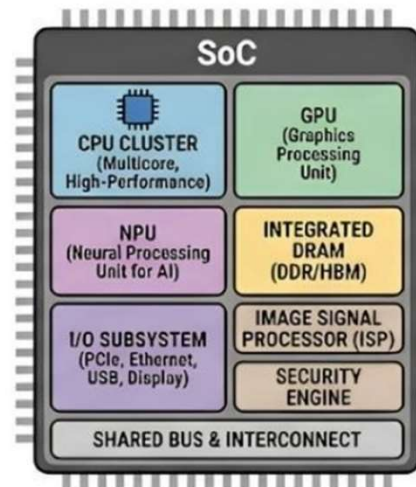
MPU

(Microprocessor Unit)
Raw CPU + External Memory



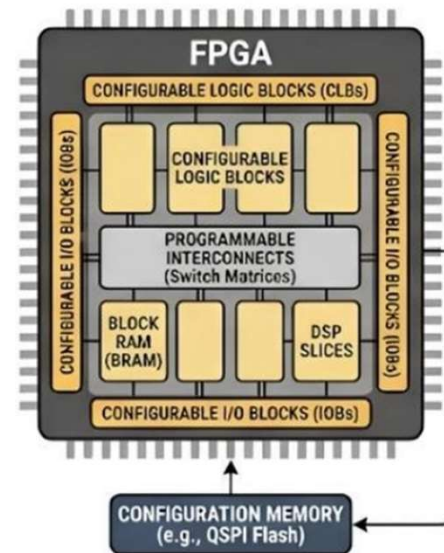
SoC

(System-on-Chip)
Highly Integrated, High-Speed



FPGA

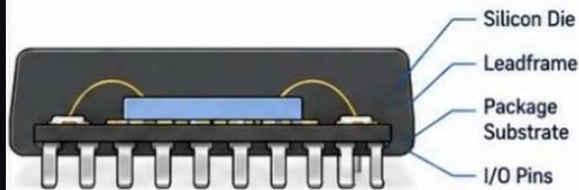
(Field-Programmable Gate Array)
Custom Programmable Logic Array



SYSTEM INTEGRATION LEVEL COMPARISON

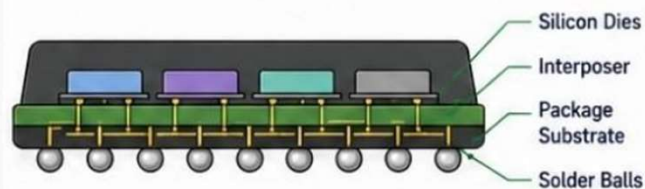
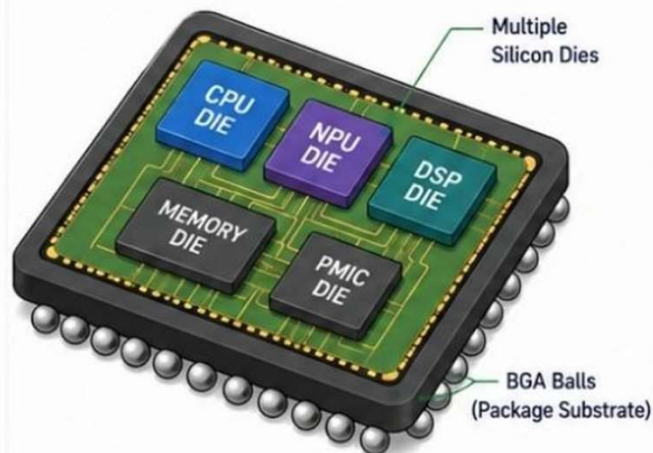
1. SoC

System on Chip



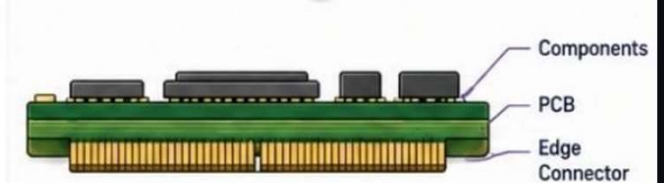
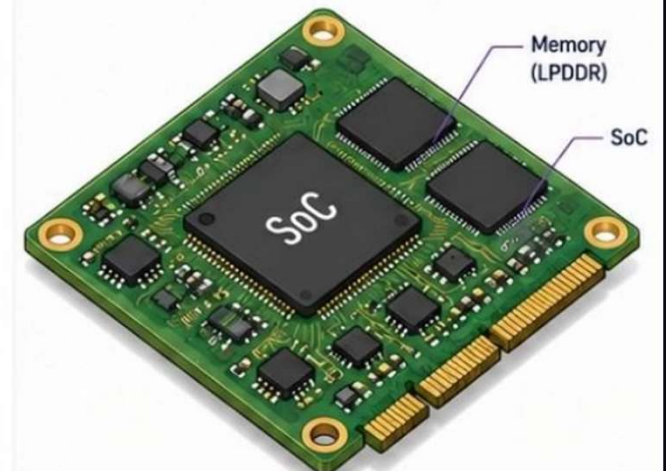
2. SiP (BGA)

Multiple Dies in One Package



3. SoM (System on Module)

Pre-Built Module (Board Level)





CPU



GPU



FPGA



ASIC



Flexibility

Efficiency

CPU vs. GPU vs. FPGA vs. ASIC

ASIC

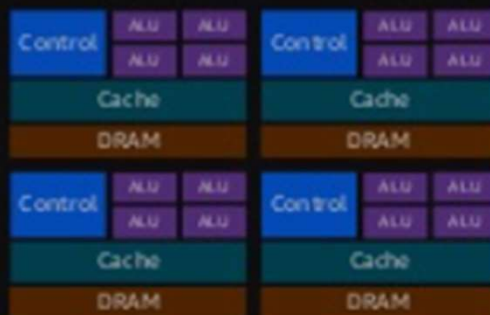
GPU



costs

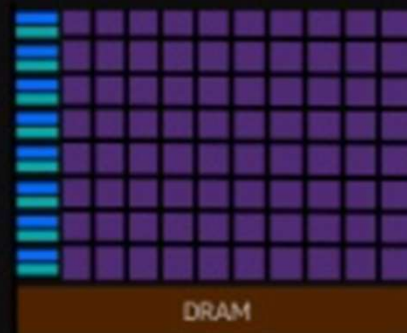
PARALLEL PROCESSING IN GPU AND FPGA

CPU



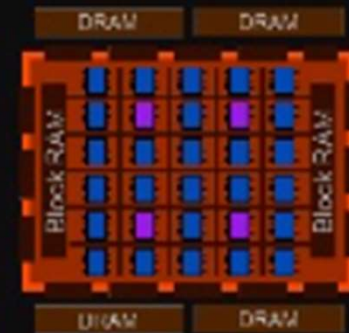
- Tens to hundreds of processing cores
- Pre-defined instruction set and datapath widths
- Optimized for general purpose computing

GPU



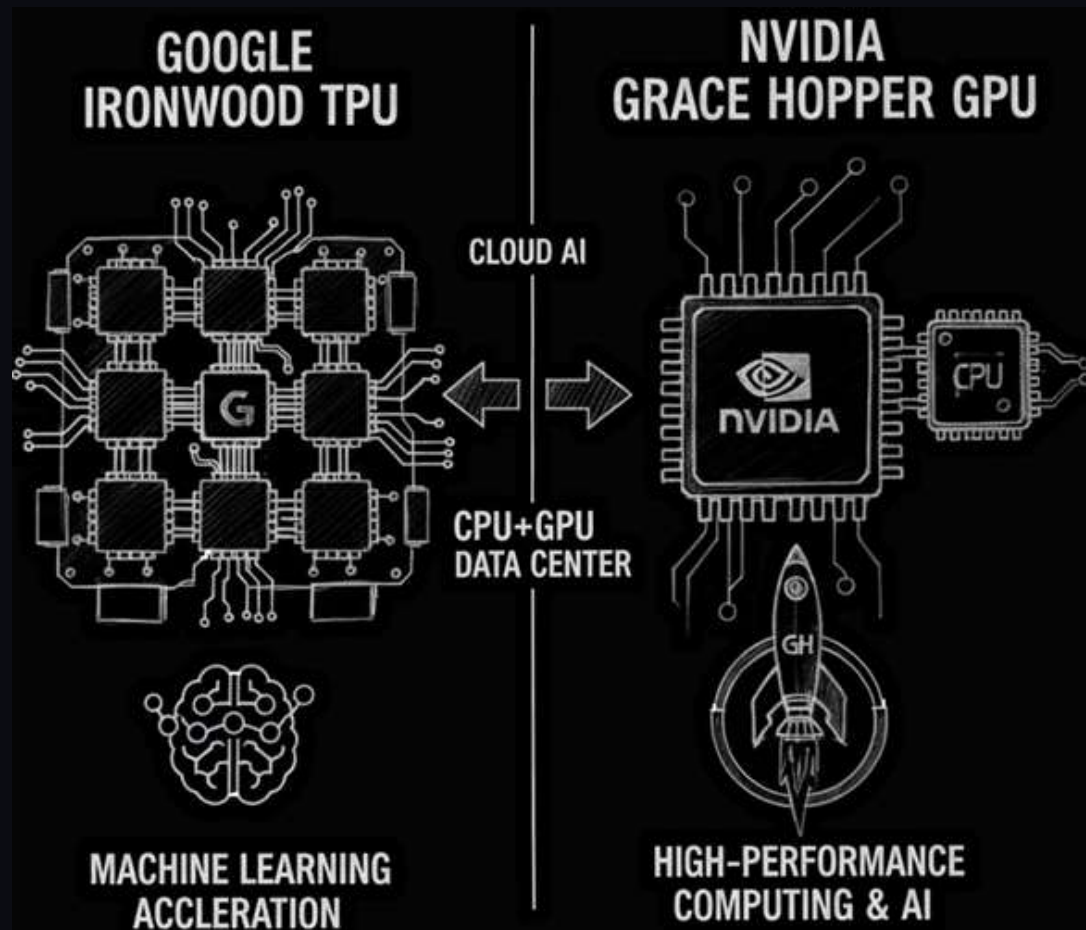
- Thousands of processing cores
- Pre-defined instruction set and datapath widths
- Highly effective at parallel execution

FPGA



- Millions of programmable digital logic cells
- No predefined instruction set or datapath widths
- Hardware-timed execution, massively parallel

TPU : Accelerate AI-Server



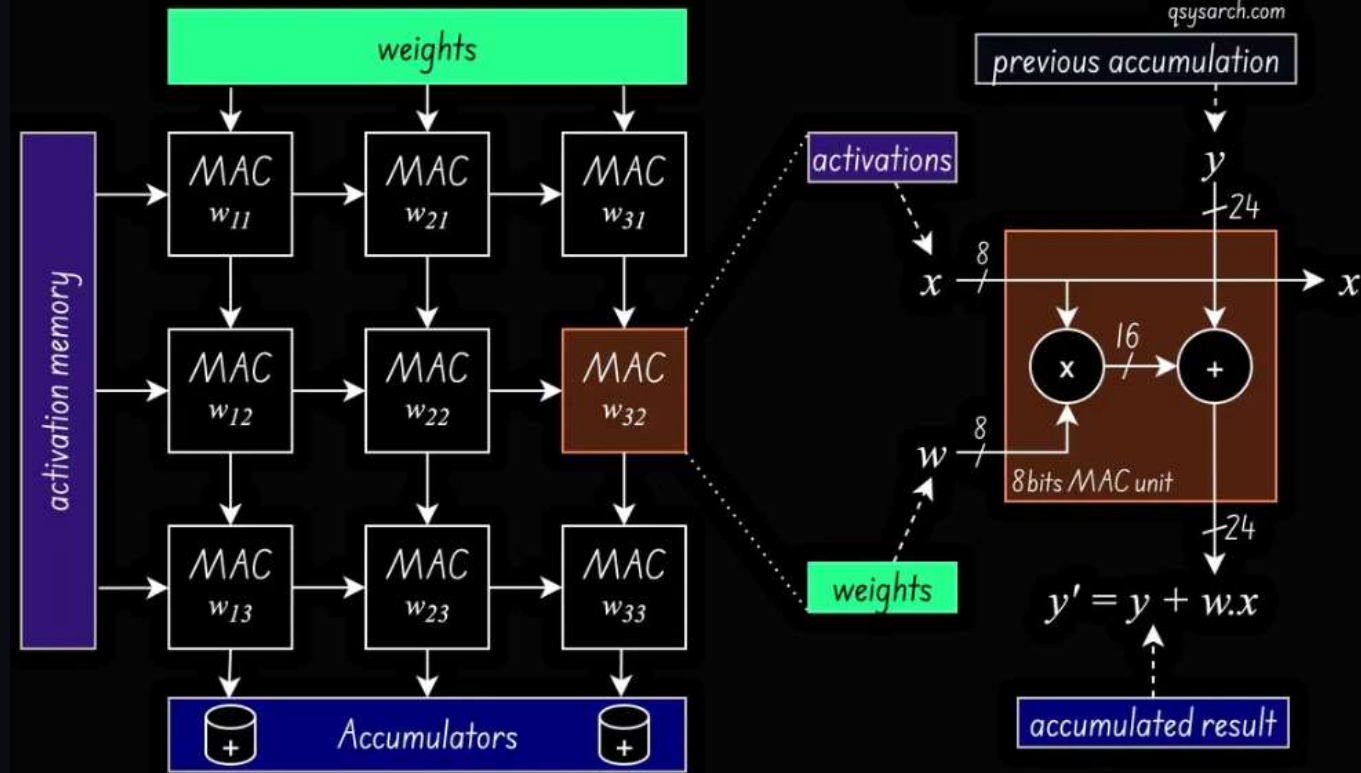
Why need Tensor Unit

Basic AI model is Neural Network

$$Y = F(\text{Input-}x * \text{Weight-}w)$$

→ many times

Architecture of a systolic array of 8-bit multiply accumulate units

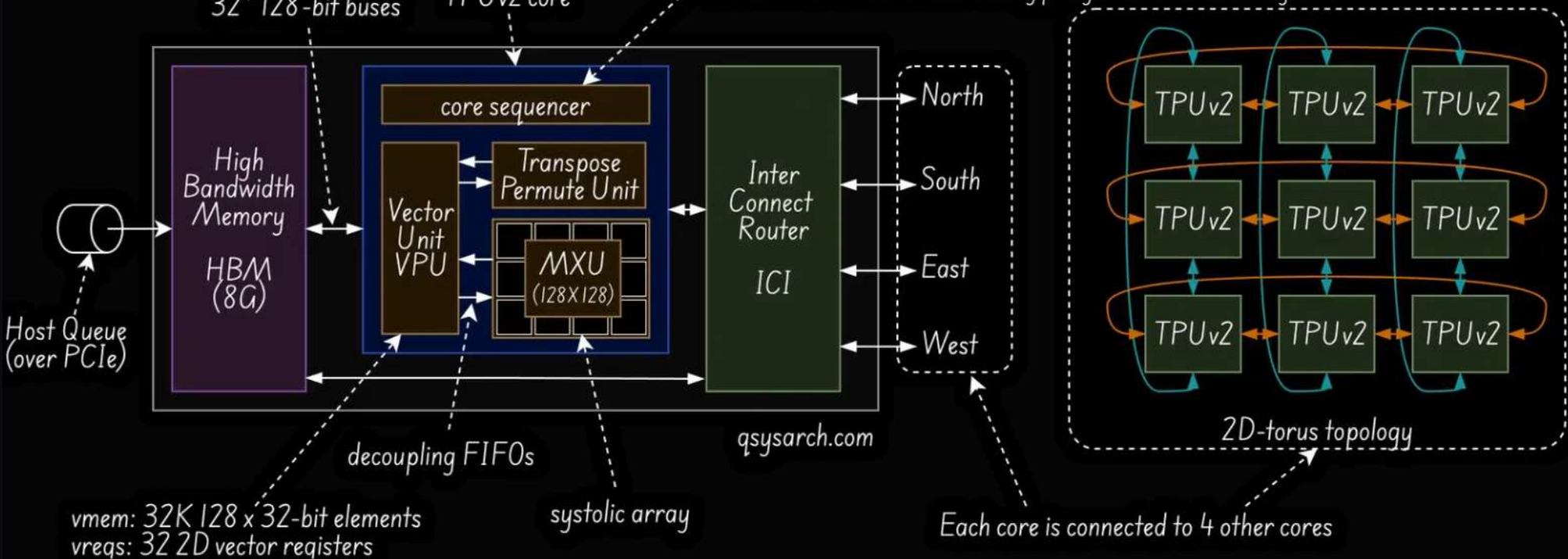


Google TPU v2 system architecture

32 * 128-bit buses

TPUv2 core

fetches 322-bit VLIW instructions and forward to the units
Smem: 4K 32-bit scalar data memory | Sregs: 32 32-bit scalar registers





DATA-CENTER AI ACCELERATORS



Large models



High throughput



HBM



Accelerator clusters



High-speed networking



Advanced cooling



EDGE AI ACCELERATORS



Low power



Low latency



Local processing



Compact accelerators



Limited memory

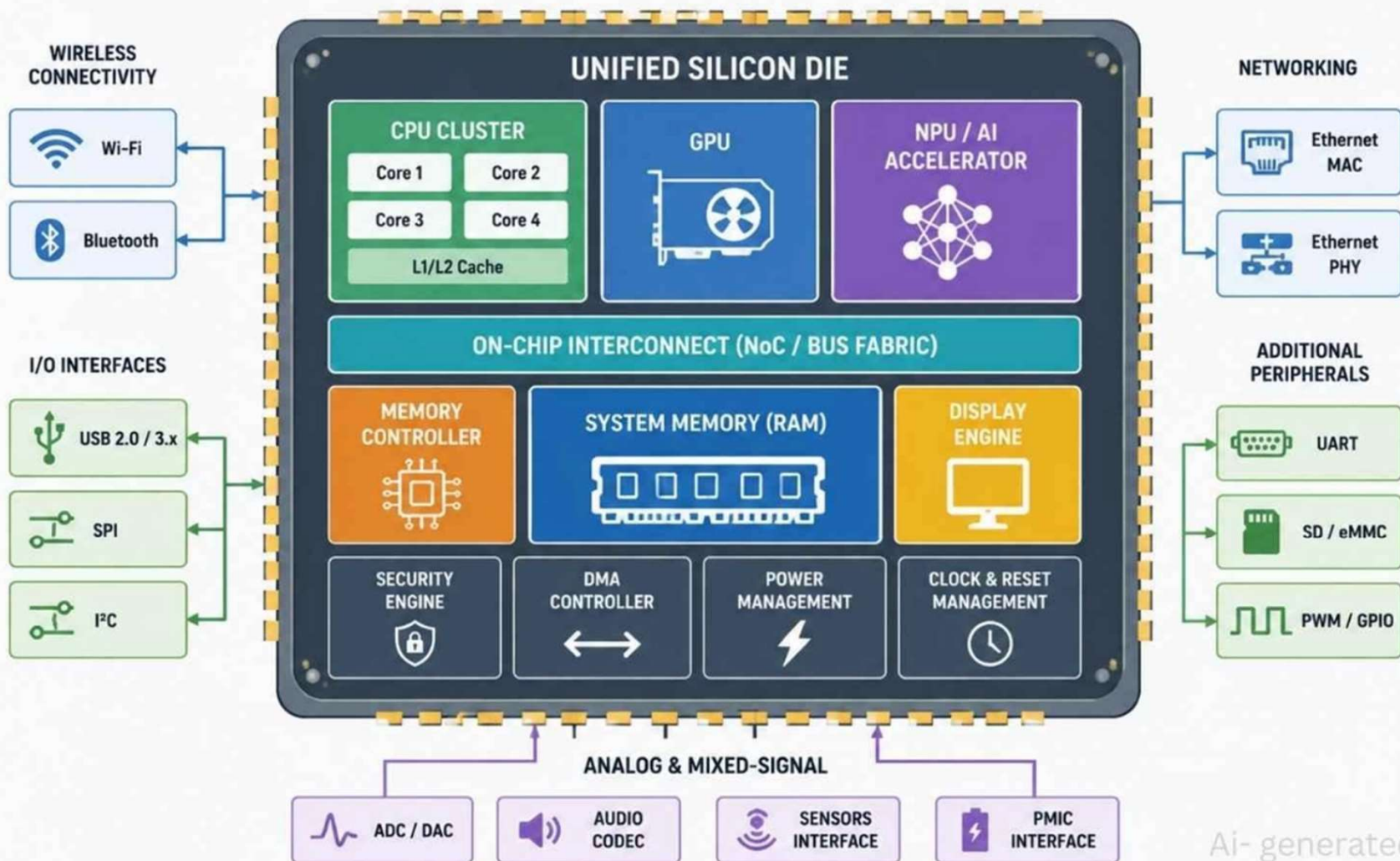


Offline capable



SYSTEM ON CHIP (SoC) ARCHITECTURE

All key components integrated on a single silicon die



Ai-generated